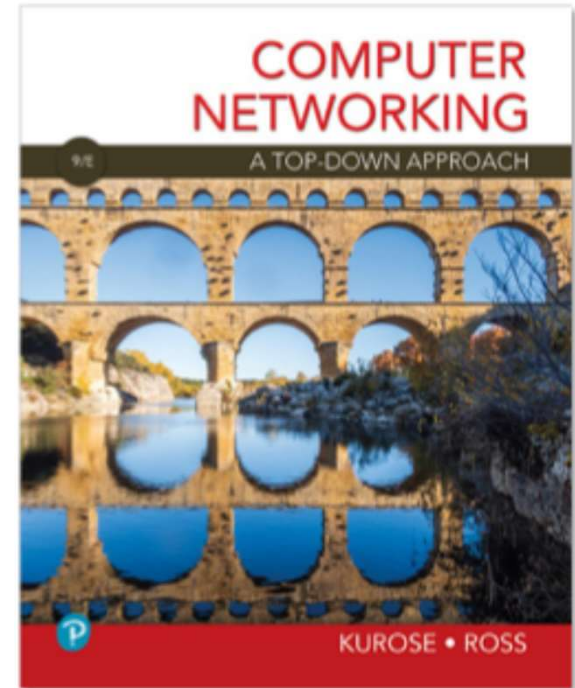


The Link Layer and LANs



Computer Networking: A Top-Down Approach

9th edition

Jim Kurose, Keith Ross

Pearson, 2025

Core Terminology (Layer-2 View)

1 What is a Node?

A node is any device that runs a Link-Layer (Layer 2) protocol.

Examples of nodes:

- Hosts (PC, laptop, server)
- Routers
- Switches
- Wi-Fi Access Points

 Key idea:

If a device participates in framing, MAC addressing, or link-layer transmission, it is a node.

2 What is a Link?

A link is a communication channel between two adjacent nodes.

Examples of links:

- Ethernet cable
- Wi-Fi radio connection
- Fiber-optic cable
- Copper wire

 Important:

- A link connects only two neighboring (adjacent) nodes
- End-to-end communication uses many links, not one

3 How Data Moves End-to-End

To send a datagram from source host → destination host:

- The datagram does NOT travel in one hop
- It moves link by link
- On each link, the datagram is:
 - Encapsulated into a frame
 - Transmitted
 - Decapsulated at the next node

Example (from the text):

Wireless Host → Server path passes through 6 links:

1. Wi-Fi link (Host → Access Point)
2. Ethernet (AP → Switch)
3. Ethernet (Switch → Router)
4. Router-to-router link
5. Ethernet (Router → Switch)
6. Ethernet (Switch → Server)



 Even though IP address stays same,

 MAC addresses and frames change at every link

4 Encapsulation at the Link Layer

Over each link:

- The network-layer datagram (IP packet) is wrapped inside:
 - A link-layer frame
- Different links → different frame formats:
 - Wi-Fi frame
 - Ethernet frame
 - Fiber frame

 This is why Layer 2 is called hop-by-hop delivery

Functions of the Link Layer (Layer 2)

The Link Layer is responsible for moving data hop-by-hop between adjacent nodes over a single link.

The main functions are:

1. Framing
2. Link Access (MAC)
3. Reliable Delivery (optional)
4. Error Detection & Correction

1 Framing

What is Framing?

Framing means:

Encapsulating a network-layer datagram (IP packet) inside a link-layer frame before transmission.

Why Framing is Needed

- Physical links transmit bits, not IP packets
- Receiver must know:
 - Where a packet starts
 - Where it ends
- Headers provide:
 - Addressing
 - Control information
 - Error checking

Frame Structure (General)

less

 Copy code

| Header | Data (IP Datagram) | Trailer |

- Header: MAC addresses, control info
- Data: IP packet
- Trailer: Error detection (CRC)

📌 Different link-layer protocols → different frame formats:

- Ethernet frame
- Wi-Fi frame
- PPP frame

Exam One-liner

Framing is the process of encapsulating a network-layer datagram into a link-layer frame for transmission over a link.

2 Link Access (Medium Access Control - MAC)

What is Link Access?

Defines:

Who can transmit and when on a shared medium.

Two Cases

✓ Point-to-Point Links

- One sender
- One receiver
- Example: PPP link
- No contention, so MAC is simple or unnecessary

✗ Broadcast (Shared) Links

- Multiple nodes share same medium
- Example:
 - Ethernet
 - Wi-Fi
- Collision can occur

👉 This is called the Multiple Access Problem

MAC Protocol Role

- Coordinates transmissions
- Prevents or resolves collisions

Examples of MAC Protocols

- CSMA/CD (Ethernet)
- CSMA/CA (Wi-Fi)
- TDMA
- FDMA

Exam One-liner

A MAC protocol controls access to a shared communication medium and coordinates frame transmission among multiple nodes.

3 Reliable Delivery (Optional)

What is Reliable Delivery?

Guarantees:

Each frame is delivered without error across a single link.

How It Works

- Acknowledgments (ACKs)
- Retransmissions (ARQ)
- Sequence numbers

Similar to TCP, but:

- Operates hop-by-hop
- Not end-to-end

Where It Is Used

✓ Wireless links

- High bit-error rate
- Local retransmission saves time

✗ Wired links

- Very low error rate
- Reliable delivery is unnecessary overhead

✚ Therefore:

- Ethernet → No reliable delivery
- Wi-Fi → Reliable delivery present

🧠 Exam One-liner

Link-layer reliable delivery ensures error-free frame transmission across a single link using acknowledgments and retransmissions.

4 Error Detection and Correction

Why Errors Occur

- Signal attenuation
- Electromagnetic interference
- Noise

Result:

- Bit flips (0 → 1 or 1 → 0)

Error Detection

Detects:

Whether an error has occurred

How:

- Sender adds error-detection bits
- Receiver checks consistency

Common Techniques:

- Parity check
- Checksum
- CRC (most common)

✚ Implemented in hardware, fast and efficient

Error Correction

Does:

Detects and fixes bit errors

How:

- Extra redundant bits
- Identify exact error location

Example:

- Hamming Code

✚ More overhead than detection

✚ Used in noisy links

Detection vs Correction

Feature	Detection	Correction
Detects errors	✓	✓
Fixes errors	✗	✓
Overhead	Low	High
Speed	Fast	Slower

Link layer, LANs: roadmap

- LANs
 - addressing, ARP
 - switches
 - VLANs
- link virtualization: MPLS



When A needs to send data to B,
it goes through switch. But
how to know which it goes through?

IP address $\rightarrow$ virtual, not portable.
Switch does not know IP Address.

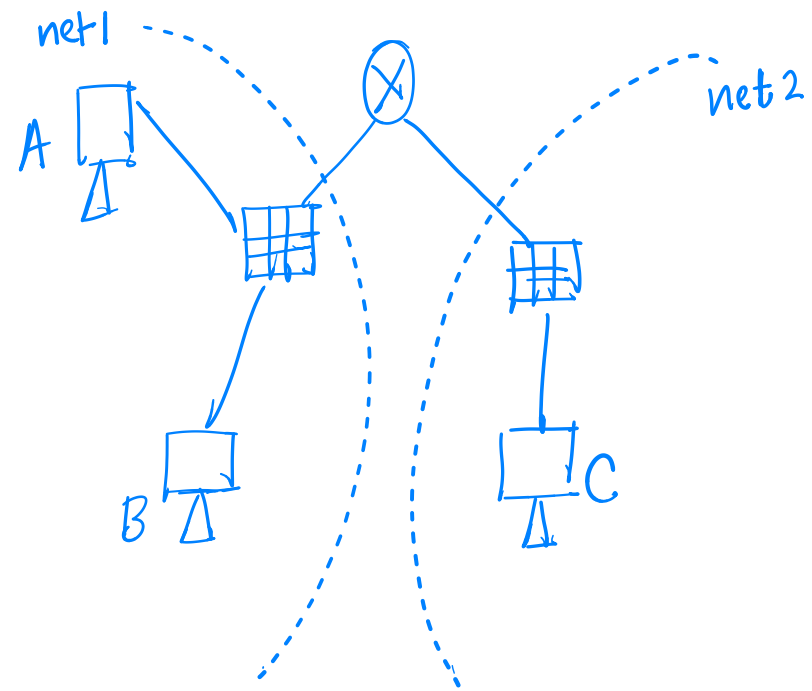
Solⁿ: MAC address.

$\hookrightarrow$ needed for efficiency

$\hookrightarrow$ does not change with network.

Router $\rightarrow$ calculates the path

Switch $\rightarrow$ only knows MAC address.



MAC addresses

- 32-bit IP address:
 - *network-layer* address for interface
 - used for layer 3 (network layer) forwarding
 - e.g.: 128.119.40.136
- MAC (or LAN or physical or Ethernet) address:
 - function: used “locally” to get frame from one interface to another physically-connected interface (same subnet, in IP-addressing sense)
 - 48-bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - e.g.: 1A-2F-BB-76-09-AD
 - hexadecimal (base 16) notation*
(each “numeral” represents 4 bits)

1 Who Actually Has a MAC Address?

Important correction to a common misconception:

- ✗ Hosts and routers do NOT have MAC addresses
- ✓ Network interfaces (adapters / NICs) have MAC addresses

Examples:

- A laptop with:
 - Ethernet NIC → 1 MAC address
 - Wi-Fi NIC → 1 MAC address
- A router with 4 interfaces → 4 MAC addresses
- One device → multiple interfaces → multiple MAC addresses

2 Why Switches Don't Need MAC Addresses on Host Ports

Key Point:

Link-layer switches operate transparently

- Hosts do not address frames to switches
- Frames are addressed only to destination adapters
- Switches simply:
 - Read frame headers
 - Forward frames accordingly

✦ Switch ports connecting to hosts do not need MAC addresses
(Though switches may have a MAC for management)

3 What Is a MAC Address?

A MAC address is:

- Link-layer address
- Also called:
 - Physical address
 - LAN address

MAC Address Properties:

Property	Value
Length	6 bytes (48 bits)
Total possible	2^{48}
Representation	Hexadecimal
Example	4A:30:10:21:10:1A

4 MAC Address Format

A 6-byte MAC address:

ruby

Copy code

(AA:BB:CC:DD:EE:FF)

Each pair:

- 1 byte 8 bits
- Written in hexadecimal

5 How MAC Addresses Are Guaranteed Unique

Managed by IEEE

Process:

- Company buys 24-bit prefix (OUI - Organizationally Unique Identifier)
- IEEE fixes first 24 bits
- Manufacturer assigns last 24 bits

css

Copy code

[IEEE Assigned (24 bits)][Manufacturer Assigned (24 bits)]

✦ This ensures:

No two adapters in the world have the same MAC address

6 Are MAC Addresses Permanent?

✦ Originally: Yes

Now:

- MAC addresses can be changed via software
- Used for:
 - Privacy
 - Virtual machines
 - Security testing

✦ For theory/exams:

Assume MAC addresses are fixed

7 Flat vs Hierarchical Addressing (VERY IMPORTANT)

MAC Address: Flat

- No structure
- No location information
- Does not change when moving networks

IP Address: Hierarchical

- Network part + host part
- Changes when switching networks

💡 Analogy

Real World	Network
Social Security Number	MAC Address
Postal Address	IP Address

📌 You need both:

- MAC → local delivery
- IP → global delivery

8 How Frames Are Delivered Using MAC Addresses

Sending Side:

1. Sender inserts destination MAC address
2. Frame transmitted on LAN

Receiving Side:

1. Adapter checks destination MAC
2. If match:
 - Extract IP datagram
 - Pass to Network Layer
3. If not:
 - Discard frame silently

📌 CPU is interrupted only if MAC matches



9 Broadcast MAC Address

Sometimes:

A sender wants everyone on the LAN to receive the frame

Broadcast MAC Address:

ruby

Copy code

FF:FF:FF:FF:FF:FF

- 48 consecutive 1s
- Used by:
 - ARP
 - DHCP
 - Network discovery

📌 All adapters must accept and process broadcast frames

10 Summary Table (Exam Gold)

Feature	MAC Address
Layer	Link Layer
Assigned to	Network Interface
Length	48 bits
Structure	Flat
Global uniqueness	Yes
Changes with network	No
Used for	Local delivery
Broadcast address	FF:FF:FF:FF:FF:FF

🔥 Common Viva Traps

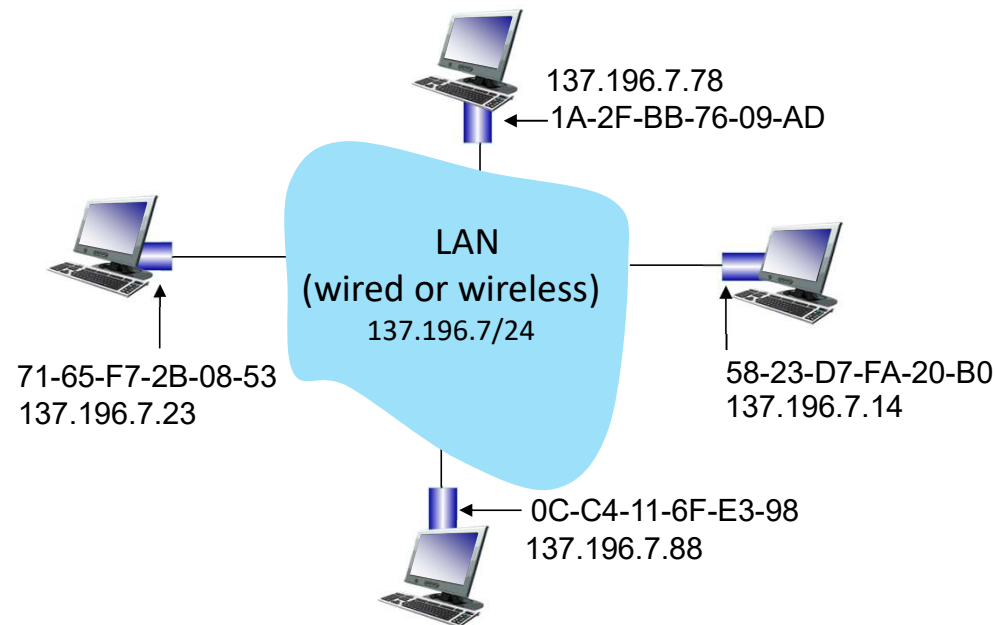
- ? Does a switch have a MAC address?
👉 Not for forwarding frames
- ? Does a router have one MAC address?
👉 No, one per interface
- ? Can two devices have same MAC?
👉 No (in theory)
- ? Why not use MAC for routing?
👉 Flat structure → not scalable



MAC addresses

each interface on LAN

- has unique 48-bit MAC address
- has a locally unique 32-bit IP address (as we've seen)

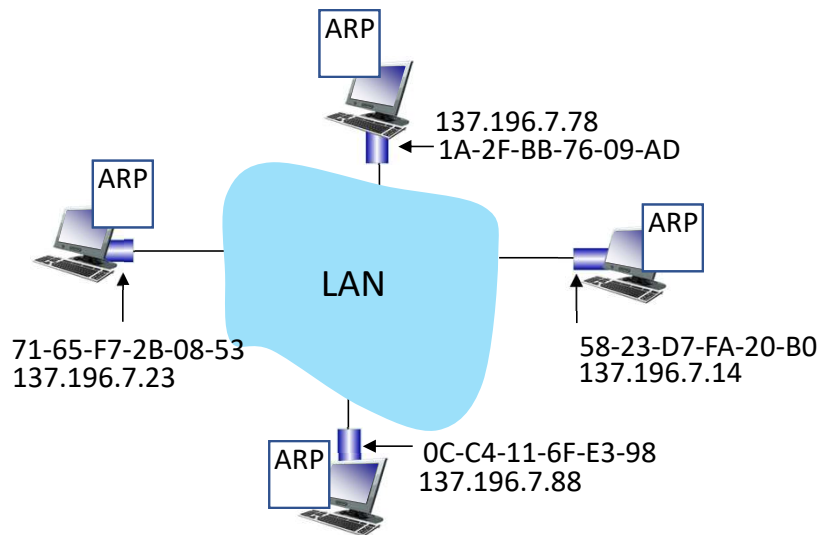


MAC addresses

- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- analogy:
 - MAC address: like Social Security Number
 - IP address: like postal address
- MAC flat address: portability
 - can move interface from one LAN to another
 - recall IP address *not* portable: depends on IP subnet to which node is attached

ARP: address resolution protocol

Question: how to determine interface's MAC address, knowing its IP address?



ARP table: each IP node (host, router) on LAN has table

- IP/MAC address mappings for some LAN nodes:
< IP address; MAC address; TTL >
- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)

1 Why ARP Is Needed

In a network, two kinds of addresses coexist:

Layer	Address
Network Layer	IP address
Link Layer	MAC address

- ✦ IP addresses are used for logical routing
- ✦ MAC addresses are used for actual frame delivery on a LAN

⚠ Problem:

To send a frame on a LAN, a host must know the destination MAC address, but applications and IP only know the destination IP address.

✓ ARP solves this problem

2 What ARP Does (One-Line Definition)

ARP maps (resolves) an IP address to its corresponding MAC address on the same subnet.

3 Where ARP Works (Very Important)

- ARP works only within the same subnet
- It cannot resolve remote IP addresses
- ✦ If destination is on another subnet:
 - ARP resolves gateway/router's MAC, not the final host

4 Example Network Scenario

Assume:

- Host A
 - IP: 222.222.222.220
 - MAC: 1A-2B-3C-4D-5E-6F
- Host B
 - IP: 222.222.222.222
 - MAC: 49-BD-D2-C7-56-2A
- Both are on the (same LAN)

🎯 Goal:

Host A wants to send an IP datagram to Host B

5 The Core Question

How does Host A know Host B's MAC address if it only knows its IP?

✓ Answer: Using ARP

6 ARP Table

Each host/router maintains an ARP table:

IP Address	MAC Address	TTL
222.222.222.222	49-BD-D2-C7-56-2A	20 min

Key Properties:

- Stored in memory
- Entries are temporary
- Typical lifetime: ~20 minutes
- Built automatically (plug-and-play)

7 ARP Resolution Process (Step-by-Step)

Case 1: Entry Exists in ARP Table

1. Host checks ARP table
 2. Finds MAC address
 3. Sends frame directly
- ✓ Fast, no ARP messages sent

Case 2: Entry Does NOT Exist (Important Case)

- Step 1: Create ARP Request

Host A creates an ARP query packet containing:

- Sender IP: 222.222.222.220
- Sender MAC: Host A's MAC
- Target IP: 222.222.222.222
- Target MAC: unknown

- Step 2: Broadcast the ARP Request

- Encapsulated in a link-layer frame
- Destination MAC:

ruby

Copy code

FF:FF:FF:FF:FF:FF

🗣 "Who has IP 222.222.222.222? Tell me your MAC!"

🗣️ "Who has IP 222.222.222.222? Tell me your MAC!"

- Step 3: All Hosts Receive the ARP Request
- Every adapter on the LAN receives the frame
- Each checks:

 "Is this IP mine?"

- Only Host B answers

- Step 4: ARP Reply (Unicast)

Host B sends an ARP reply:

- Contains its MAC address
- Sent directly (unicast) to Host A

🔴 Why unicast?

- Now Host B knows Host A's MAC from the request

- Step 5: ARP Table Update

Host A:

- Stores the IP-MAC mapping
- Sends the actual IP datagram inside a frame

8 Broadcast vs Unicast in ARP (Exam Favorite)

Message	Why
ARP Request	Destination MAC unknown → broadcast
ARP Reply	Sender MAC known → unicast

9 ARP vs DNS (Comparison)

Feature	ARP	DNS
Resolves	IP → MAC	Name → IP
Scope	Same subnet only	Entire Internet
Transport	Direct (no TCP/UDP)	UDP/TCP
Caching	ARP table	DNS cache

10 Why ARP Is Plug-and-Play

- No manual configuration
- Tables auto-populate
- Entries auto-expire
- Hosts joining/leaving are handled automatically

11 Where Does ARP Belong? (Classic Exam Question)

Confusion:

- Encapsulated in link-layer frame
- Contains IP and MAC addresses

Conclusion:

ARP straddles the boundary between the Link Layer and Network Layer

🔴 Best phrasing:

ARP does not fit cleanly into the layered model and operates between the link and network layers.

12 One-Page Flow Summary

sql

📄 Copy code

```
IP datagram ready
↓
Check ARP table
↓
Entry exists? → Send frame
↓
Else:
  Broadcast ARP request
  ↓
  Receive ARP reply
  ↓
  Update ARP table
  ↓
  Send frame
```

🧠 One-Line Exam Answer

ARP is a protocol used to resolve an IP address to a MAC address within the same subnet by broadcasting a request and receiving a unicast reply.

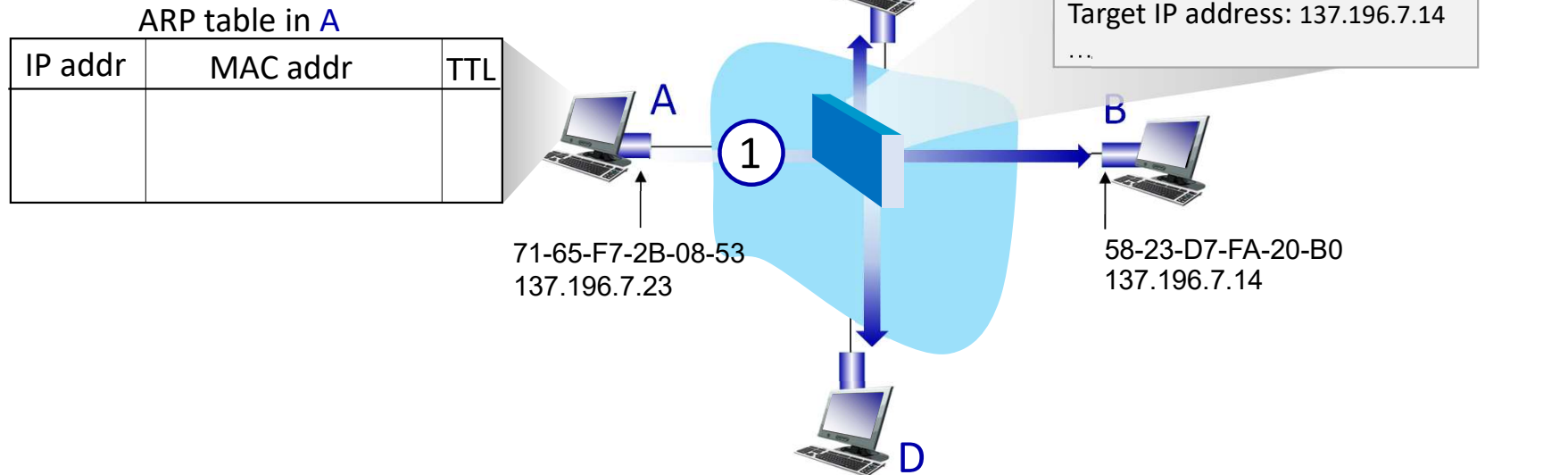
ARP protocol in action

example: A wants to send datagram to B

- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

A broadcasts ARP query, containing B's IP addr

- ①
- destination MAC address = FF-FF-FF-FF-FF-FF
 - all nodes on LAN receive ARP query

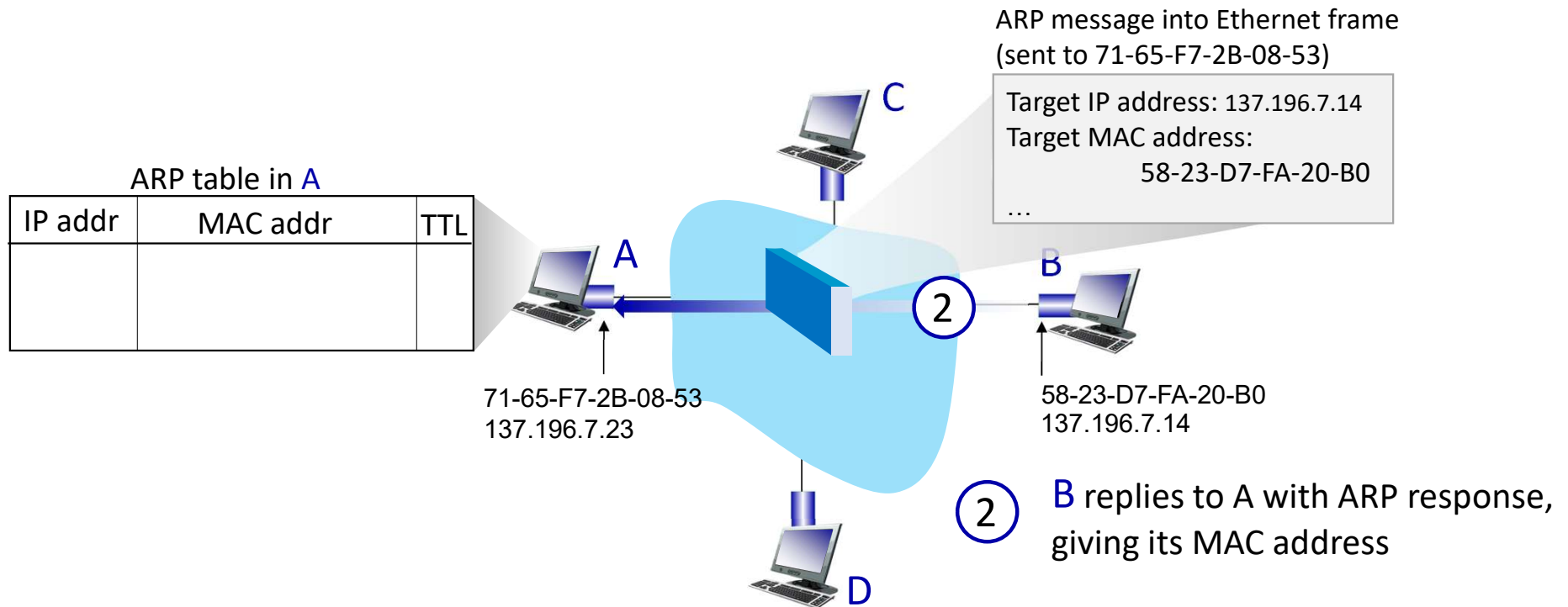


ARP protocol in action



example: A wants to send datagram to B

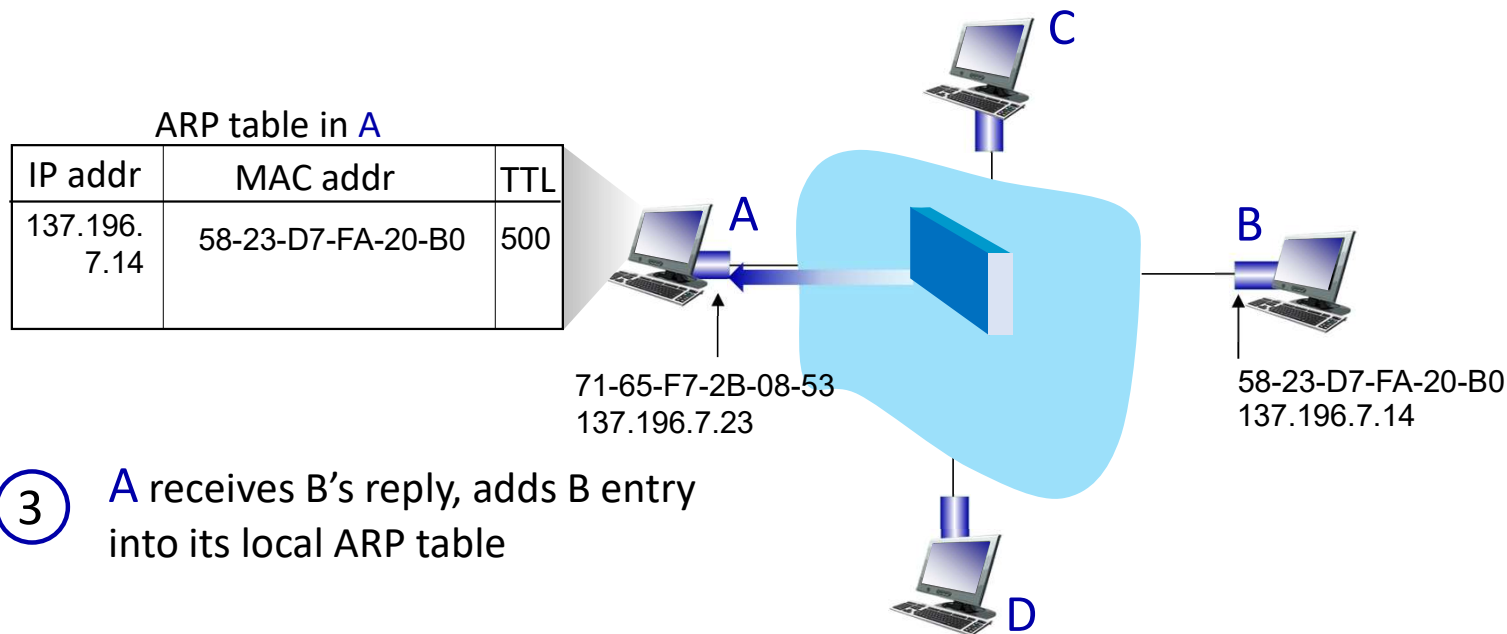
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ARP protocol in action

example: A wants to send datagram to B

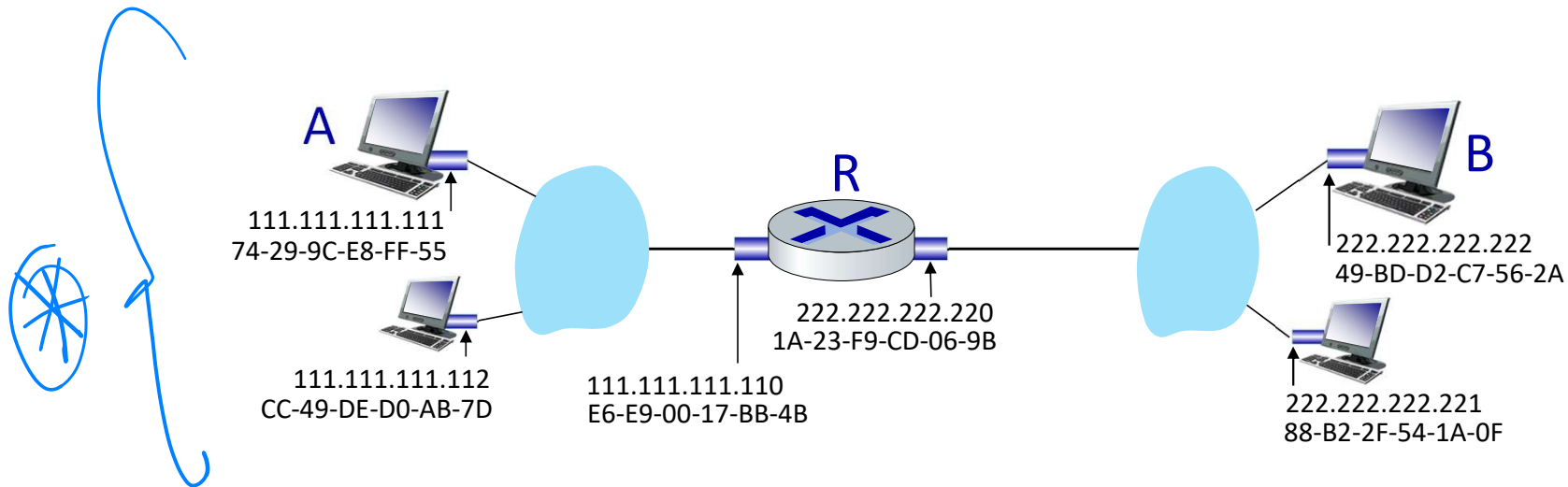
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



Routing to another subnet: addressing

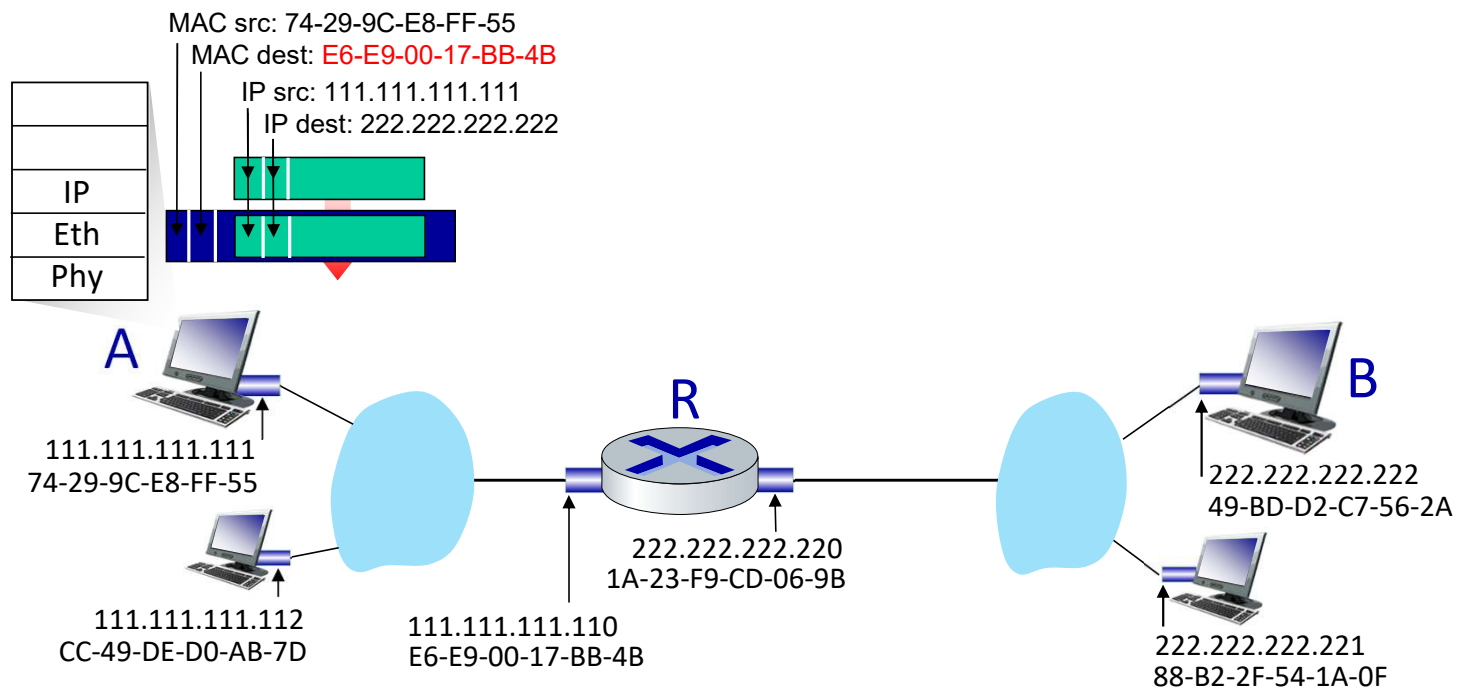
walkthrough: sending a datagram from *A* to *B* via *R*

- focus on addressing – at IP (datagram) and MAC layer (frame) levels
- assume that:
 - A knows B's IP address
 - A knows IP address of first hop router, R (how?)
 - A knows R's MAC address (how?)



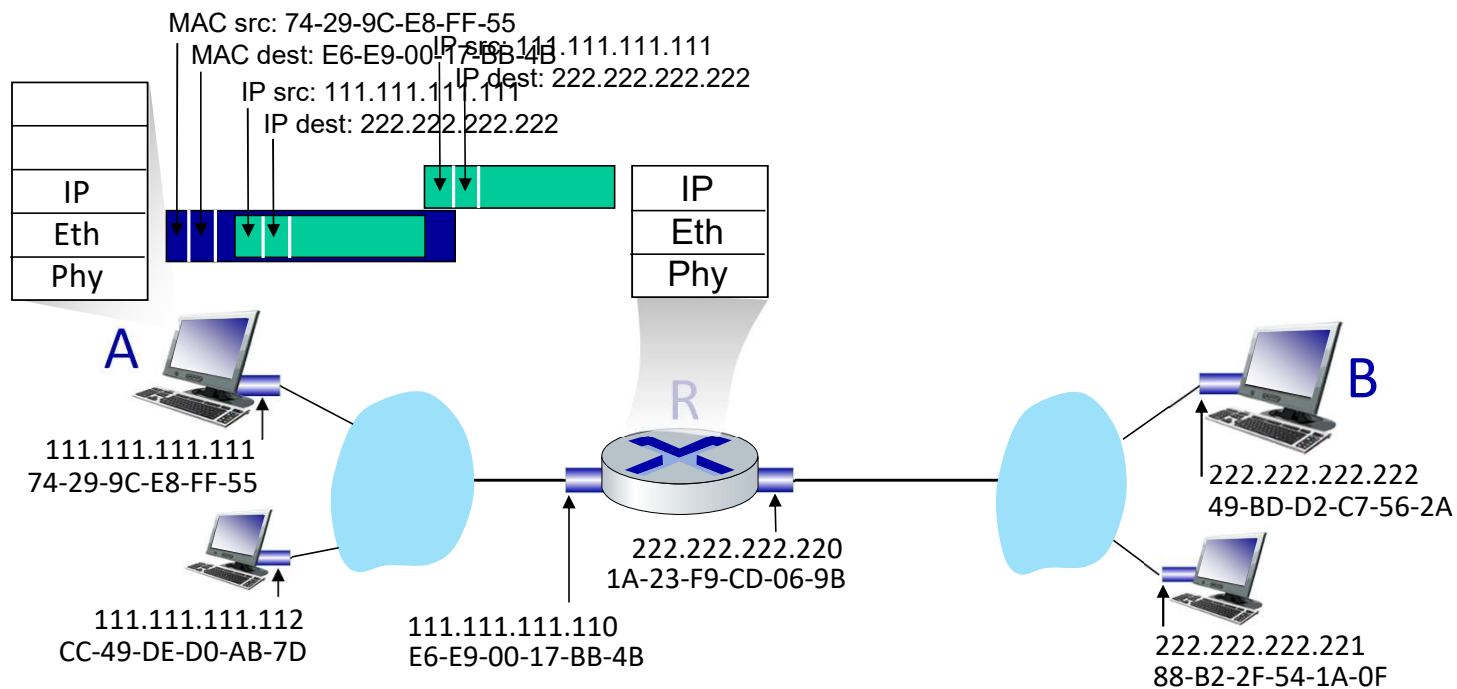
Routing to another subnet: addressing

- A creates IP datagram with IP source A, destination B
- A creates link-layer frame containing A-to-B IP datagram
 - R's MAC address is frame's destination



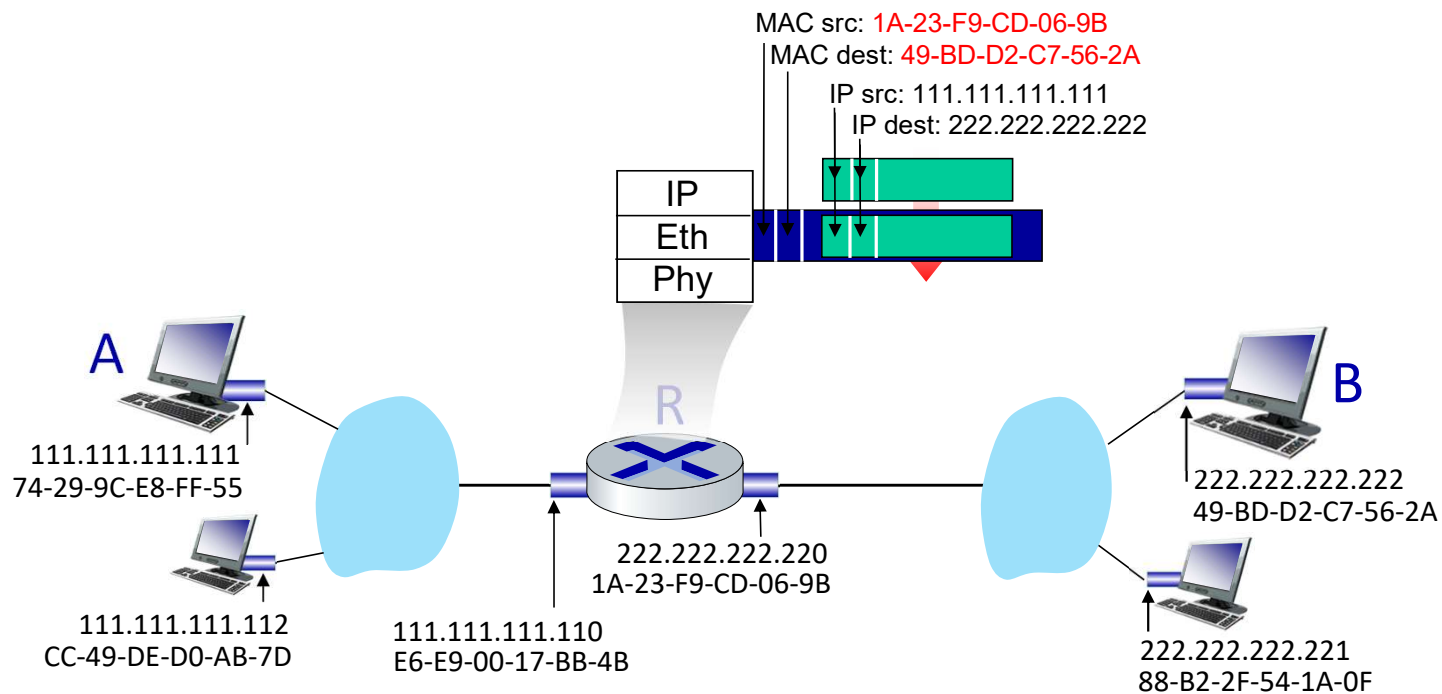
Routing to another subnet: addressing

- frame sent from A to R
- frame received at R, datagram removed, passed up to IP



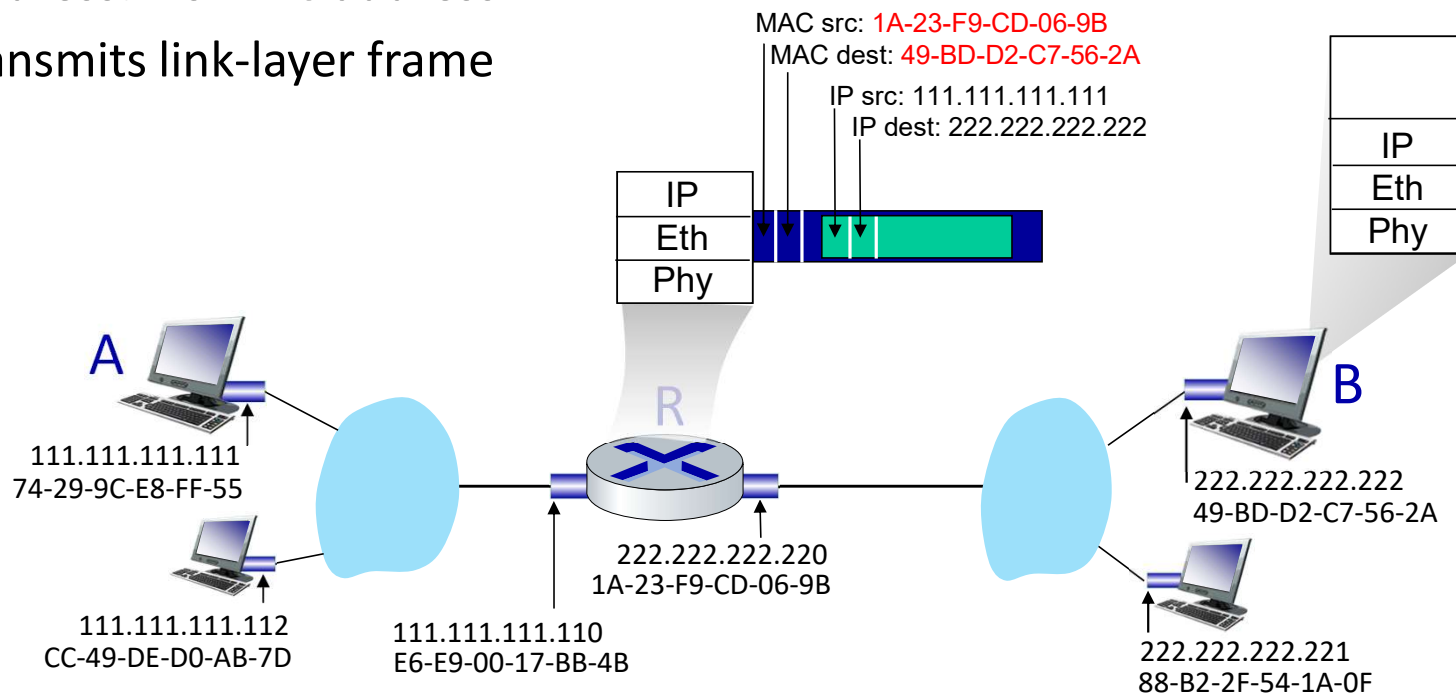
Routing to another subnet: addressing

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address



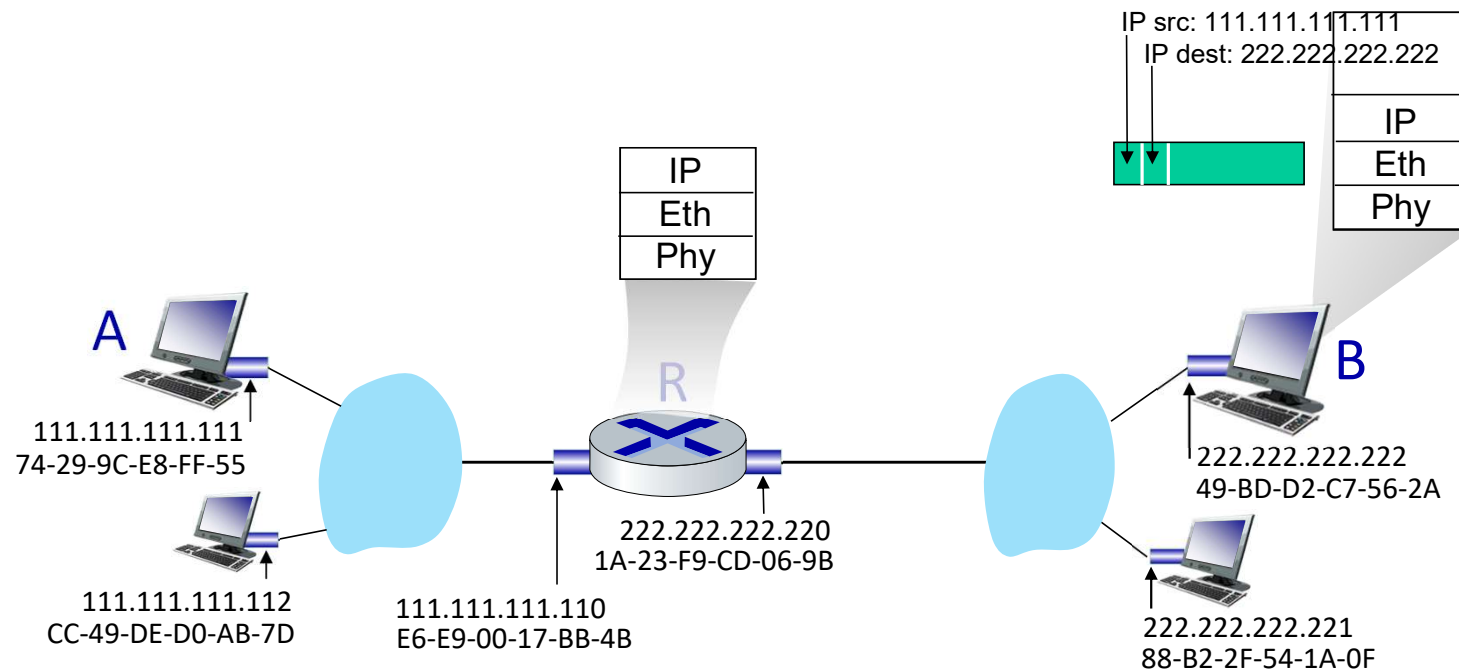
Routing to another subnet: addressing

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address
- transmits link-layer frame



Routing to another subnet: addressing

- B receives frame, extracts IP datagram destination B
- B passes datagram up protocol stack to IP



Link layer, LANs: roadmap

■ LANs

- addressing, ARP
- Ethernet
- switches
- VLANs



- a day in the life of a web request

Switch

- Switch is a **link-layer** device: takes an *active* role
 - store, forward Ethernet (or other type of) frames
 - examine incoming frame's MAC address, *selectively* forward frame to one-or-more outgoing links when frame is to be forwarded on segment, uses CSMA/CD to access segment
- **transparent**: hosts *unaware* of presence of switches
- **plug-and-play, self-learning**
 - switches do not need to be configured

1 What Is a Link-Layer Switch?

A link-layer switch is a Layer-2 device that:

- Receives Ethernet frames
- Examines destination MAC addresses
- Forwards or filters frames accordingly

📌 Key property:

A switch is transparent to hosts and routers.

This means:

- Hosts send frames directly to other hosts
- They do not address frames to the switch
- Hosts are unaware that a switch is forwarding frames

2 Why Switches Need Buffers

Sometimes:

- Frames arrive faster than an outgoing link can transmit
- This causes temporary congestion

✓ Solution:

- Each output interface has a buffer
- Similar to router output queues

3 Two Core Functions of a Switch

- 1. Filtering

Decides whether a frame should be dropped or forwarded

- 2. Forwarding

Decides to which interface(s) the frame should be sent

Both functions depend on a switch table.

4 Switch Table (MAC Table)

The switch table stores:

Field	Meaning
MAC Address	Destination device
Interface	Port leading to that MAC
Time	When entry was learned



📌 Important:

- Table may not contain all devices
- Entries expire after inactivity
- Built automatically

5 How a Switch Processes a Frame

Assume:

- Frame arrives on interface x
- Destination MAC = DD-DD-DD-DD-DD-DD

The switch checks its switch table

6 Three Possible Cases (VERY IMPORTANT)

✓ Case 1: Destination MAC NOT in switch table

- Action:
- Broadcast frame to all interfaces except x

📌 Reason:

- Switch does not know where destination is

🗣️ This is called flooding

✓ Case 2: Destination MAC maps to same interface x

- Meaning:
- Frame came from the same LAN segment
- Destination already received it

- Action:
- Discard the frame

✓ This is filtering

✓ Case 3: Destination MAC maps to different interface y ≠ x

- Meaning:
- Switch knows exactly where destination is
- Action:
- Forward frame only to interface y

✓ No broadcasting needed

7 Example from the Diagram

Suppose switch table contains:

MAC Address	Interface
62-FE-F7-11-89-A3	1
7C-BA-B2-B4-91-10	2

Example 1:

Frame destined for 62-FE-F7-11-89-A3 arrives on interface 1

- ✓ Destination already on same segment
- ✓ Switch filters (drops) frame

Example 2:

Frame destined for 62-FE-F7-11-89-A3 arrives on interface 2

- ✓ Destination reachable via interface 1
- ✓ Switch forwards frame to interface 1

8 Why Switches Are Smarter Than Hubs

Hub	Switch
Broadcasts everything	Selective forwarding
No MAC table	Uses switch table
High collisions	No collisions
Inefficient	Efficient

✓ Switches reduce unnecessary traffic

9 Switch vs Router (Quick Comparison)

Feature	Switch	Router
Layer	Layer 2	Layer 3
Address used	MAC	IP
Table	Switch (MAC) table	Routing table
Configuration	Self-learning	Routing protocols

◆ Switches vs Routers

1 Basic Difference

Aspect	Switch	Router
OSI Layer	Layer 2 (Data Link)	Layer 3 (Network)
Address Used	MAC address	IP address
Type	Layer-2 packet switch	Layer-3 packet switch
Forwards	Frames	Packets (datagrams)

📌 Both are store-and-forward devices, but they operate at different layers.

2 Modern Reality (Important Concept)

Although switches and routers are fundamentally different, modern devices blur the line:

- Modern switches (e.g., OpenFlow switches) can:
 - Forward using MAC addresses
 - Forward using IP addresses
 - Match on multiple header fields (L2, L3, L4)

This is called generalized forwarding (match + action).

3 Why Choose Between Switches and Routers?

In a network like Figure 6.15, an administrator could use either:

- A switch to interconnect LANs
- Or a router to interconnect LANs

Each choice has pros and cons.

4 Advantages of Switches

✓ Plug-and-Play

- No manual configuration required
- MAC tables are self-learned
- Very attractive for administrators

✓ High Speed

- Process frames only up to Layer 2
- Lower per-packet processing delay
- Higher forwarding rate than routers

✓ Traffic Isolation

- Frames forwarded only where needed
- No collisions (unlike hubs)

5 Disadvantages of Switches

✗ Spanning Tree Restriction

- To avoid broadcast loops:
 - Switches must form a spanning tree
 - Some links remain unused
- Cannot always use the best path

✗ Large ARP Overhead

- Large switched networks need:
 - Large ARP tables
 - Frequent ARP broadcasts
- Increases processing and traffic

✗ Broadcast Storms

- If one host floods broadcast frames:
 - Switches forward all broadcasts
 - Entire LAN can collapse

6 Advantages of Routers

✓ No Spanning Tree Limitation

- Use hierarchical IP addressing
- Packets do not normally loop
- Can use multiple active paths

📌 This is how the Internet supports:

- Redundant links
- Rich global topology

✓ Intelligent Routing

- Chooses optimal paths
- Supports routing protocols (OSPF, BGP, RIP)

✓ Broadcast Isolation

- Routers do not forward layer-2 broadcasts
- Protect network from broadcast storms

7 Disadvantages of Routers

✗ Not Plug-and-Play

- Requires:
 - IP address configuration
 - Routing table setup

✗ Slower than Switches

- Must process packets up to Layer 3
- Higher per-packet processing delay

✗ Administrative Overhead

- More complex to manage
- Requires skilled configuration

8 Summary Table (From Table 6.1)

Feature	Hubs	Switches	Routers
Traffic isolation	✗ No	✓ Yes	✓ Yes
Plug-and-play	✓ Yes	✓ Yes	✗ No
Optimal routing	✗ No	✗ No	✓ Yes

9 When to Use Switches vs Routers?

♦ Use Switches When:

- Small networks (hundreds of hosts)
- Few LAN segments
- Need simplicity and speed
- Minimal configuration desired

📌 Example:

- Small office
- University department LAN

♦ Use Routers When:

- Large networks (thousands of hosts)
- Multiple subnets
- Need:
 - Traffic isolation
 - Broadcast control
 - Optimal routing

📌 Example:

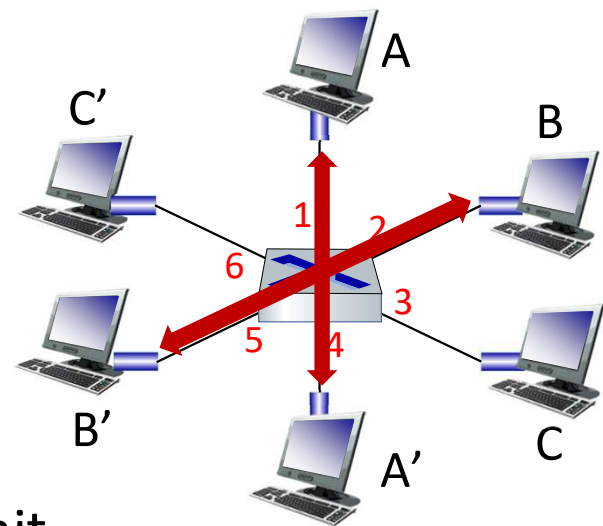
- University campus backbone
- Corporate networks
- Internet infrastructure

10 One-Line Exam Answer

Switches forward frames using MAC addresses and are fast and plug-and-play but suffer from broadcast and spanning-tree limitations, whereas routers forward packets using IP addresses, support optimal routing, isolate broadcasts, and are essential for large, scalable networks.

Switch: multiple simultaneous transmissions

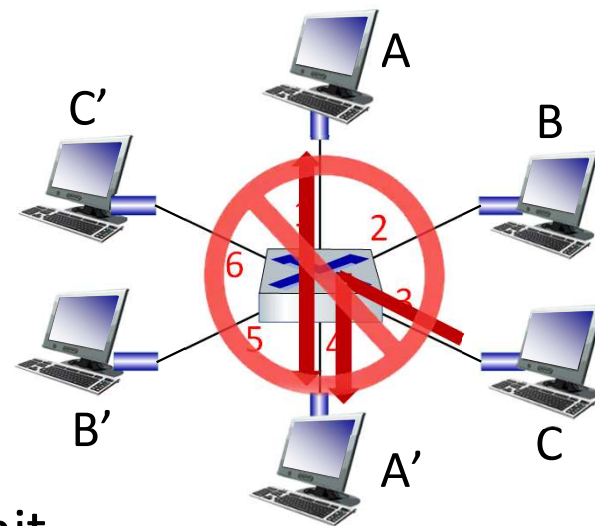
- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - no collisions; full duplex
 - each link is its own collision domain
- **switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six
interfaces (1,2,3,4,5,6)

Switch: multiple simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - no collisions; full duplex
 - each link is its own collision domain
- **switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions
 - but A-to-A' and C to A' can *not* happen simultaneously



switch with six
interfaces (1,2,3,4,5,6)

Switch forwarding table

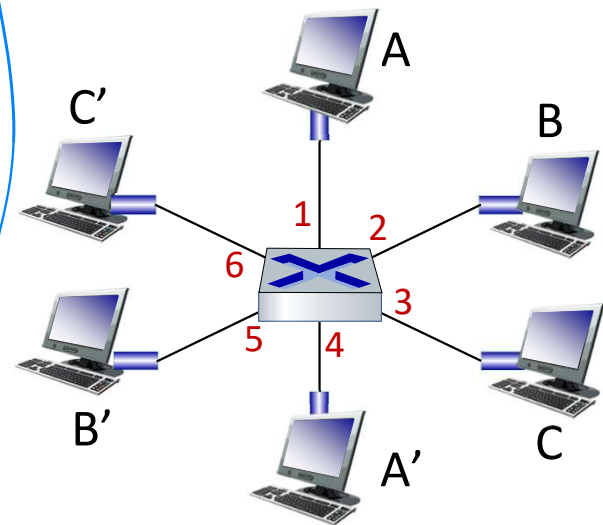
Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a **switch table**, each entry:

- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

Q: how are entries created, maintained in switch table?

- something like a routing protocol?



♦ Self-Learning in Link-Layer Switches

One of the most important and useful properties of a link-layer switch is that it is self-learning. This means the switch can automatically build and maintain its switch table without any manual configuration by a network administrator.

Because of this property, switches are plug-and-play devices.

♦ What is Self-Learning?

Self-learning is the process by which a switch dynamically learns which MAC address is reachable through which interface by observing incoming frames.

🔄 How Self-Learning Works (Step-by-Step)

Step 1: Empty Table Initially

- When a switch is powered on:
 - Its switch table is empty
 - It knows nothing about host locations

Step 2: Learning from Incoming Frames

For every frame received on an interface, the switch:

1. Reads the source MAC address from the frame
2. Records:
 - Source MAC address
 - Incoming interface
 - Current timestamp
3. Stores this information in the switch table

♦ This tells the switch:

“This MAC address lives on this interface.”

If all hosts send frames at least once, all hosts will eventually be learned.

Step 3: Aging (Automatic Cleanup)

- Each table entry has an aging time (e.g., 60 minutes)
- If no frame is received from a MAC address during this time:
 - The switch removes the entry
- This handles:
 - Host removal
 - Host replacement
 - Device relocation

🧠 Example (From the Text)

At Time 9:39

- Frame arrives on interface 2
- Source MAC: 01-12-23-34-45-56
- Address not in table → New entry added

MAC Address	Interface	Time
01-12-23-34-45-56	2	9:39
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36

Aging Example

- Aging time = 60 minutes
- No frame from 62-FE-F7-11-89-A3 between 9:32 and 10:32
- At 10:32, the switch deletes that entry

🔌 Why Switches Are Plug-and-Play

Because of self-learning:

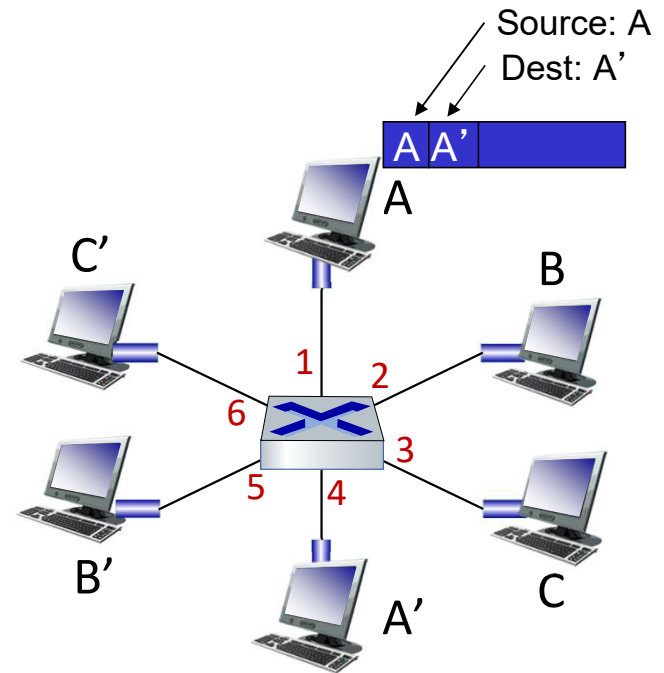
- No manual MAC table configuration
- No routing protocols needed
- No administrator intervention required

♦ To install a switch:

Just connect LAN segments to its interfaces – nothing else!

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
- when frame received, switch “learns” location of sender: incoming LAN segment
- records sender/location pair in switch table



MAC addr	interface	TTL
A	1	60

*Switch table
(initially empty)*

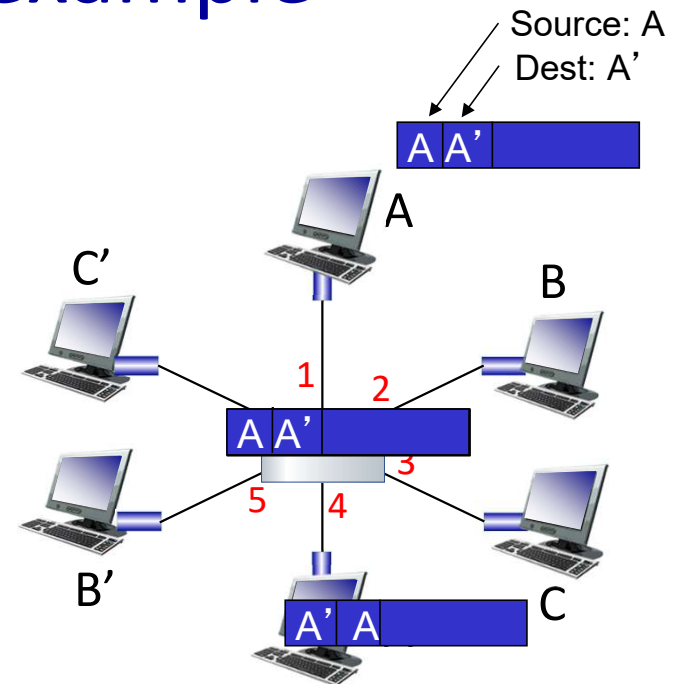
Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
then {
 if destination on segment from which frame arrived
 then drop frame
 else forward frame on interface indicated by entry
}
else flood /* forward on all interfaces except arriving interface */

Self-learning, forwarding: example

- frame destination, A',
location unknown: **flood**
- destination A location
known: **selectively send**
on just one link

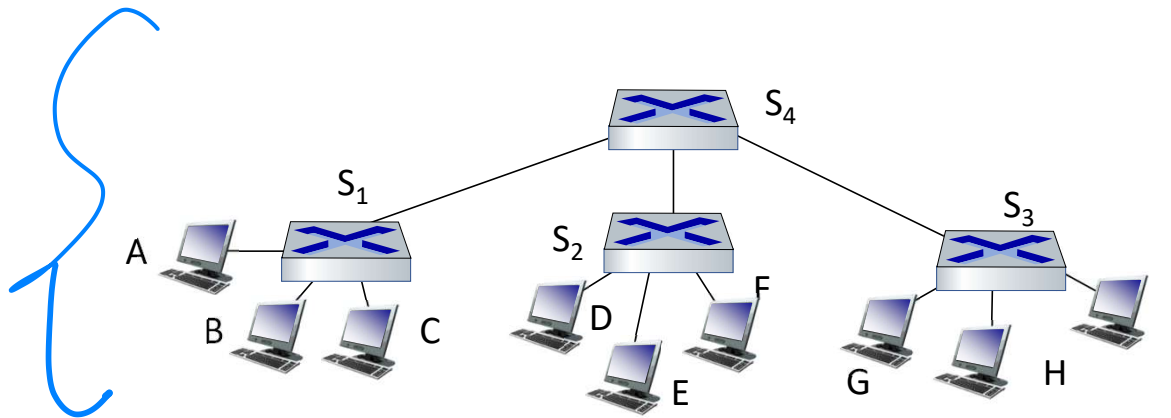


MAC addr	interface	TTL
A	1	60
A'	4	60

*switch table
(initially empty)*

Interconnecting switches

self-learning switches can be connected together:

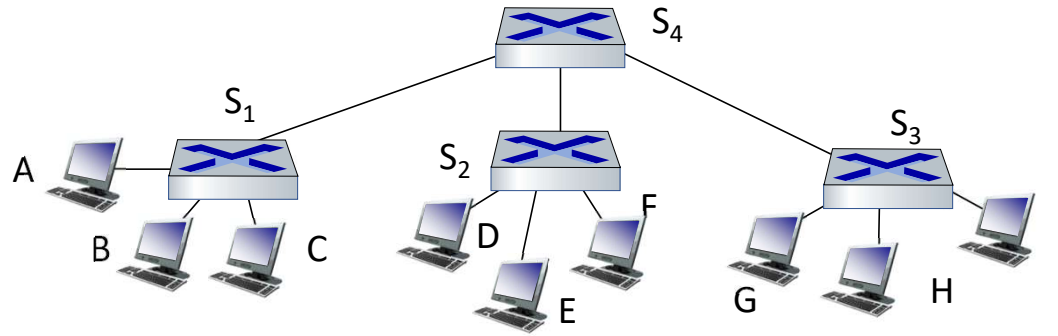


- ⌚ **Q:** sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?
- **A:** self learning! (works exactly the same as in single-switch case!)



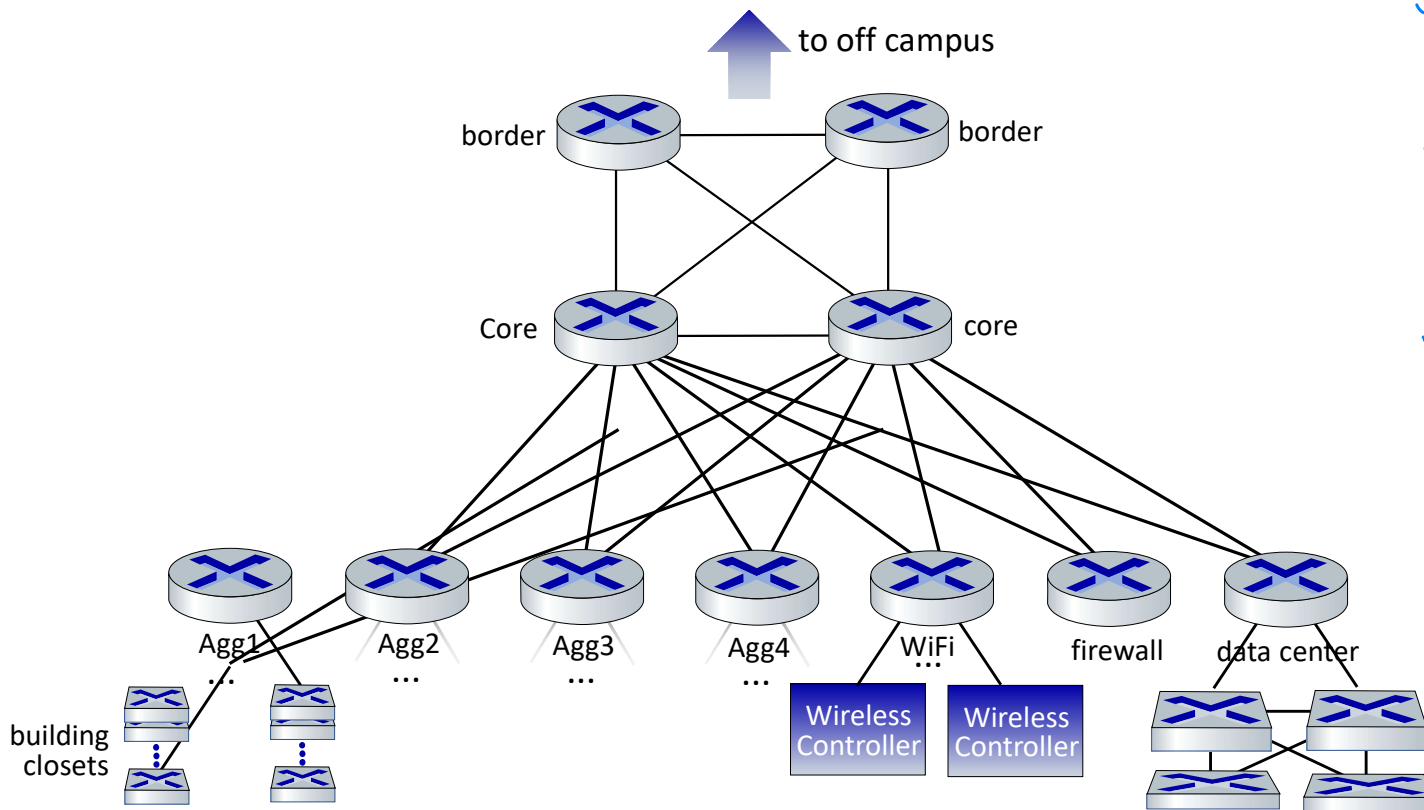
Self-learning multi-switch example

Suppose C sends frame to I, I responds to C



Q: show switch tables and packet forwarding in S₁, S₂, S₃, S₄

UMass Campus Network - Detail

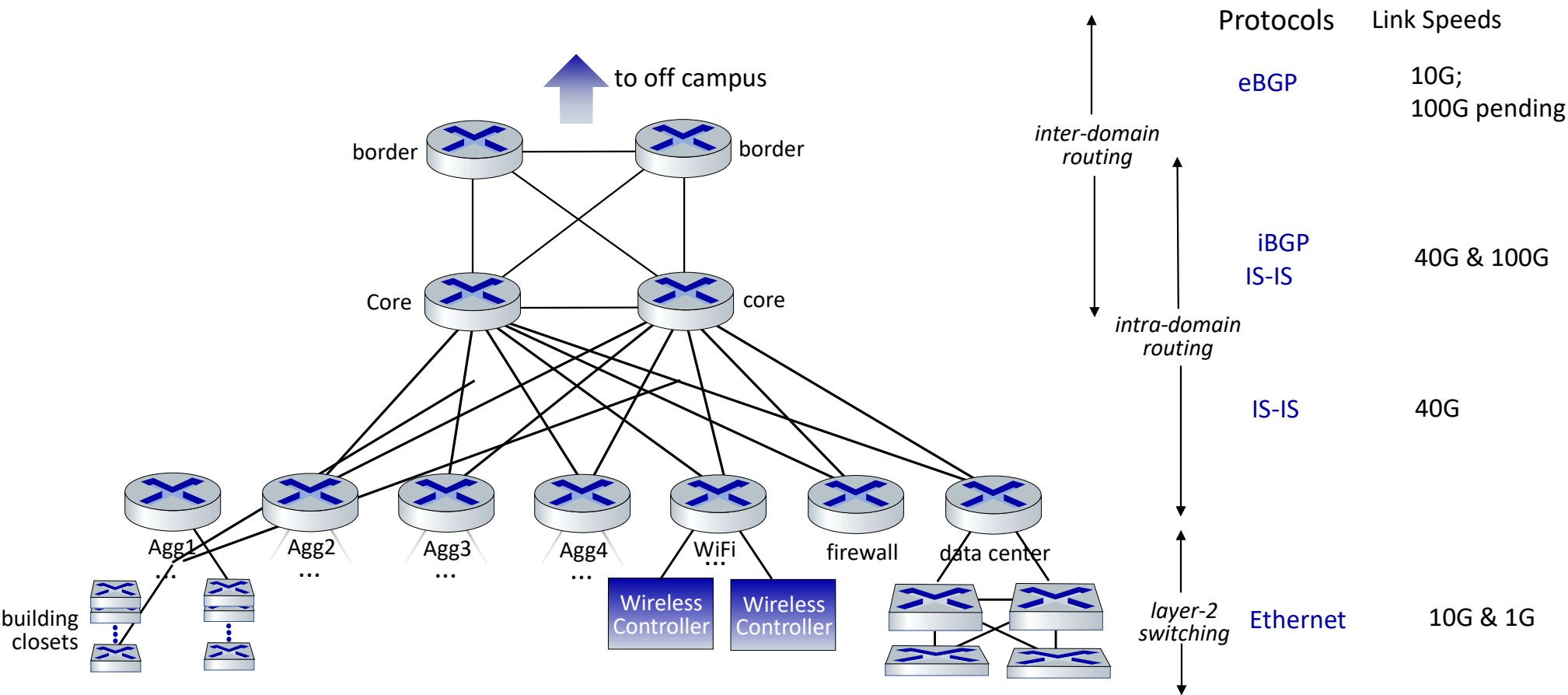


UMass network:

- 4 firewalls
- 10 routers
- 2000+ network switches
- 6000 wireless access points
- 30000 active wired network jacks
- 55000 active end-user wireless devices

... all built,
operated,
maintained by ~15
people

UMass Campus Network - Detail



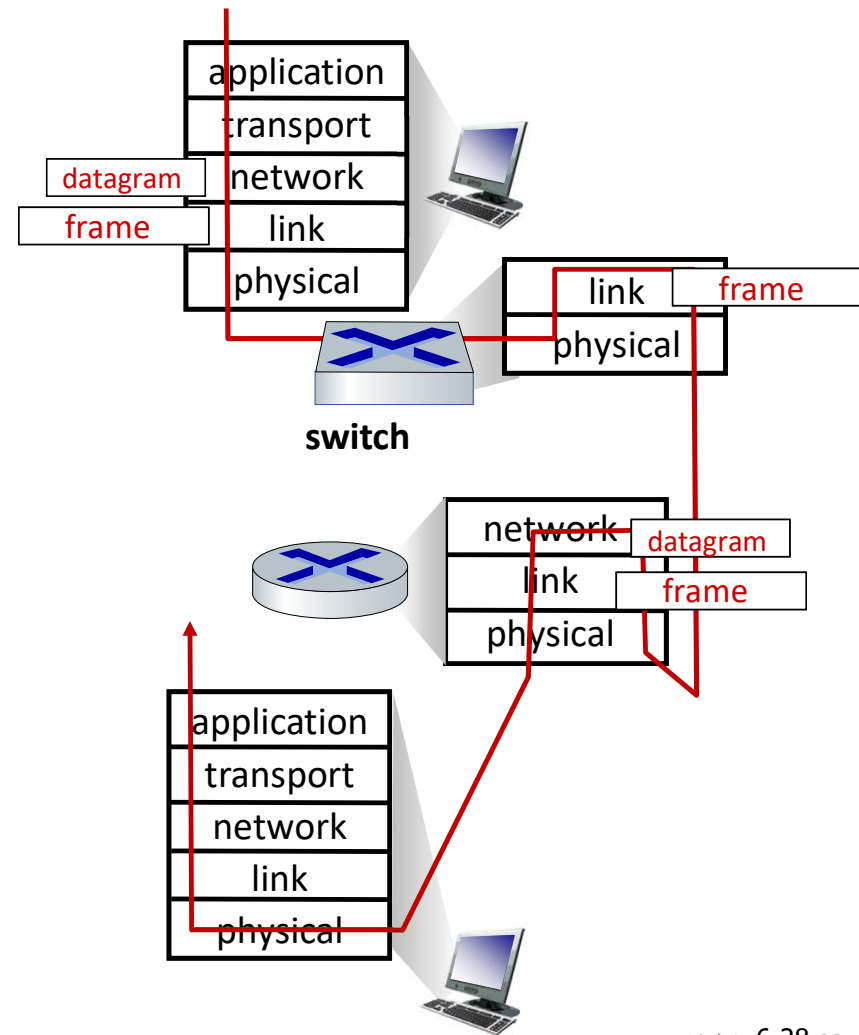
Switches vs. routers

both are store-and-forward:

- **routers:** network-layer devices (examine network-layer headers)
- **switches:** link-layer devices (examine link-layer headers)

both have forwarding tables:

- **routers:** compute tables using routing algorithms, IP addresses
- **switches:** learn forwarding table using flooding, learning, MAC addresses



Link layer, LANs: roadmap

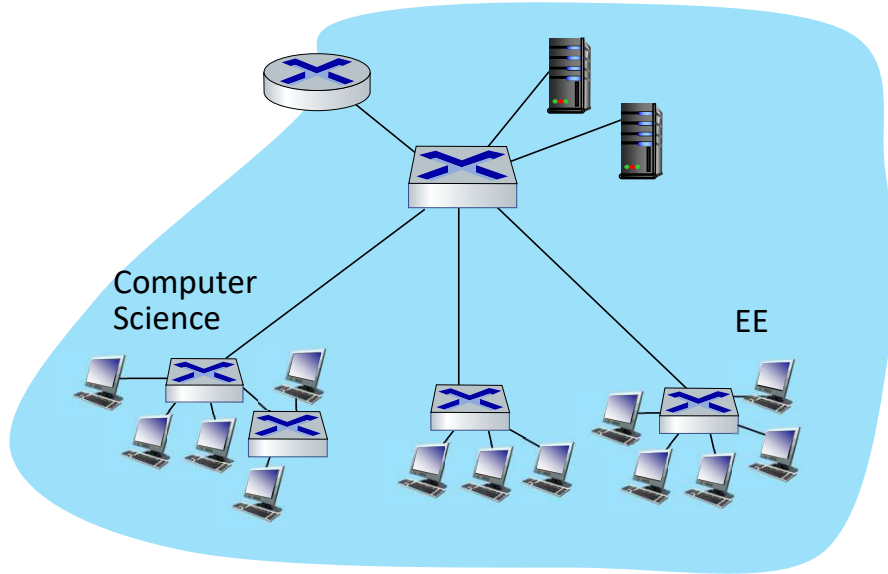
- LANs
 - addressing, ARP
 - switches
 - VLANs
- link virtualization: MPLS





Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?



single broadcast domain:

- *scaling*: all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy issues

♦ Virtual Local Area Networks (VLANs)

A VLAN (Virtual Local Area Network) allows a single physical switch infrastructure to be divided into multiple logical LANs, providing traffic isolation, flexibility, and better resource utilization.

✗ Problems with Traditional Switched LANs

In a hierarchical switched LAN (like Figure 6.15), three major problems arise:

1 Lack of Traffic Isolation

- Broadcast traffic (ARP, DHCP, unknown MAC frames) spreads across the entire LAN
- Causes:
 - Performance degradation
 - Security/privacy risks (e.g., packet sniffing)
- ♦ Using routers could solve this, but VLANs provide a layer-2 solution

2 Inefficient Use of Switches

- Each department/workgroup needs a separate switch
- Small groups waste switch ports
- One large switch cannot isolate traffic without VLANs

3 Difficult User Management

- Moving an employee between departments requires physical recabling
- Users belonging to multiple groups complicate the setup

✓ VLAN Solution

A VLAN-capable switch allows multiple virtual LANs to coexist on one physical switch.

✚ Hosts within the same VLAN:

- Communicate as if they are on the same LAN
- Are isolated from hosts in other VLANs

♦ Port-Based VLANs (Most Common)

- Switch ports are grouped into VLANs
- Each VLAN forms its own broadcast domain
- Broadcast traffic stays within the VLAN

Example

- Ports 2-8 → EE VLAN
- Ports 9-15 → CS VLAN
- Ports 1 and 16 → unassigned

✓ Solves:

- Traffic isolation
- Efficient switch usage
- Easy user movement (just reconfigure the port)

🔄 Inter-VLAN Communication

Problem

VLANs are isolated – hosts in different VLANs cannot communicate directly

Solution: Routing Between VLANs

Two approaches:

1 External Router

- A switch port connects to a router
- Router belongs to multiple VLANs
- Traffic:
 - EE VLAN → Router → CS VLAN

2 Integrated Switch + Router (Modern Solution)

- Vendors provide Layer-3 switches
- VLAN switching + routing in one device
- No external router required



VLAN Trunking (Connecting VLANs Across Switches)

When VLANs span multiple switches, simple port-to-port connections don't scale.

Poor Scalability

- N VLANs require N physical links between switches

VLAN Trunking (Scalable Solution)

- Special trunk ports connect switches
- Trunk ports:
 - Carry traffic for all VLANs
 - Use VLAN tags to identify VLAN membership

IEEE 802.1Q VLAN Tagging

To identify VLANs over trunk links, switches use 802.1Q tagging.

802.1Q Frame Format

- Standard Ethernet frame
 - 4-byte VLAN tag

VLAN Tag Fields

Field	Size	Purpose
TPID	2 bytes	Fixed value 0x8100
VLAN ID	12 bits	Identifies VLAN
Priority	3 bits	QoS (similar to IP TOS)

Tag added by:

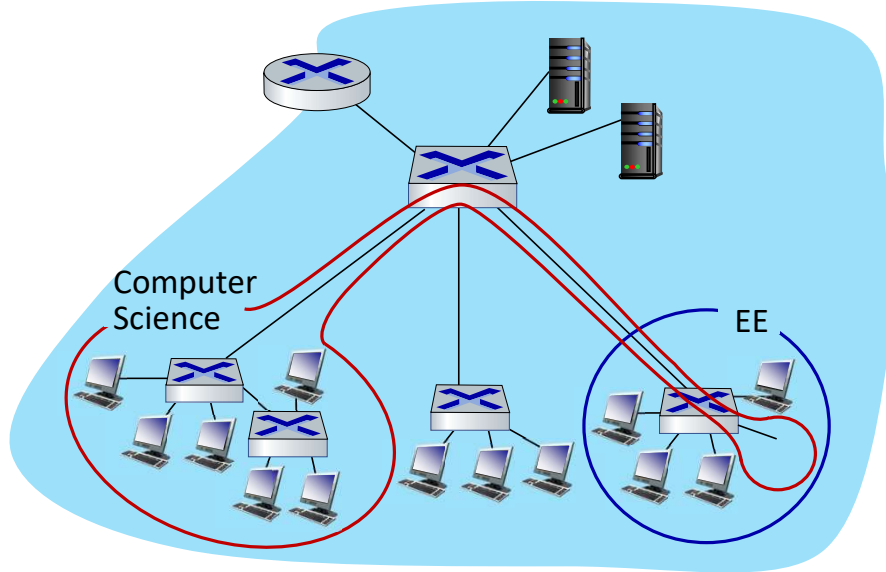
- Sending switch on trunk

Tag removed by:

- Receiving switch

Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?



single broadcast domain:

- *scaling*: all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy, efficiency issues

administrative issues:

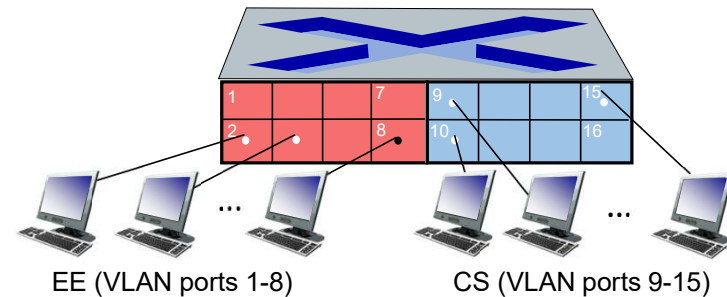
- CS user moves office to EE - *physically* attached to EE switch, but wants to remain *logically* attached to CS switch

Port-based VLANs

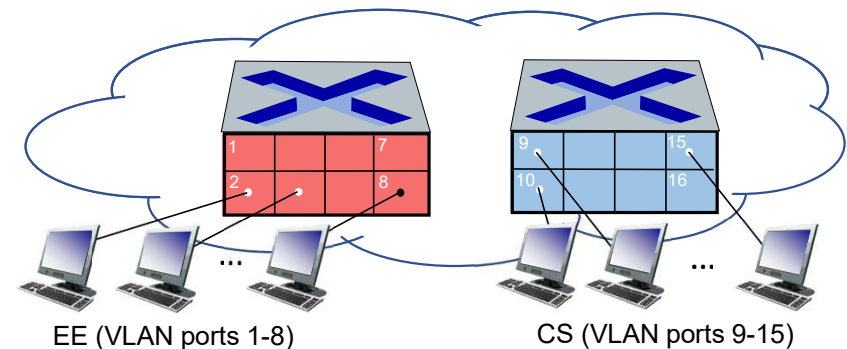
Virtual Local Area Network (VLAN)

switch(es) supporting VLAN capabilities can be configured to define multiple *virtual* LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch

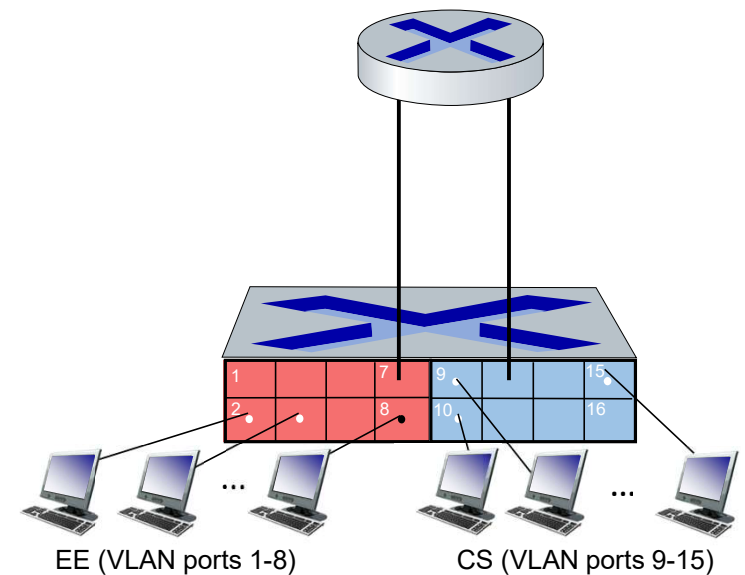


... operates as **multiple** virtual switches

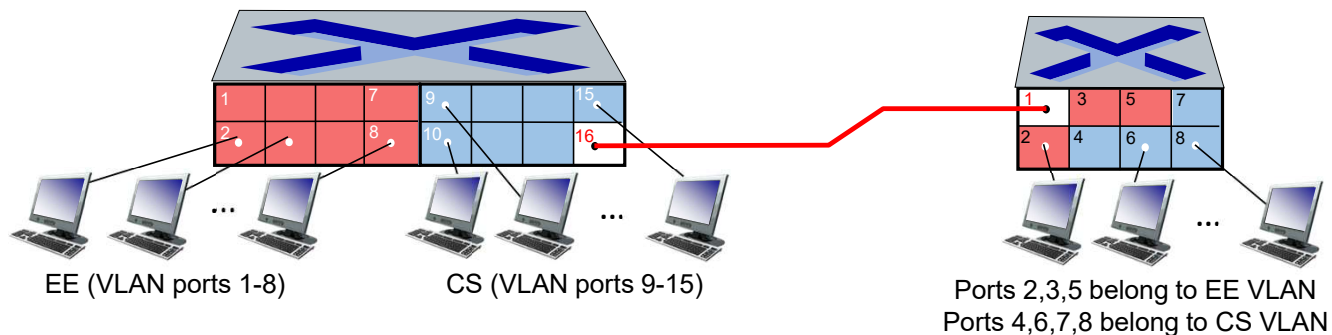


Port-based VLANs

- **traffic isolation:** frames to/from ports 1-8 can *only* reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **dynamic membership:** ports can be dynamically assigned among VLANs
- **forwarding between VLANs:** done via routing (just as with separate switches)
 - in practice vendors sell combined switches plus routers



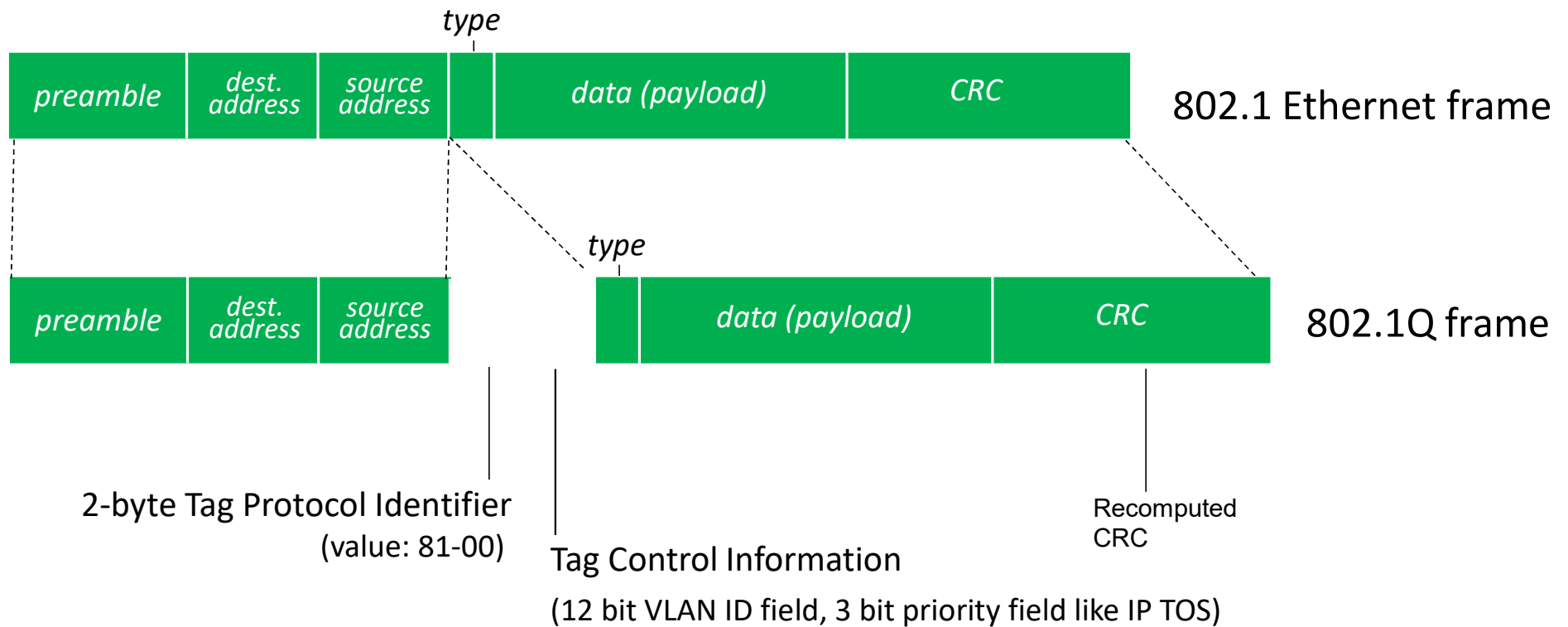
VLANs spanning multiple switches



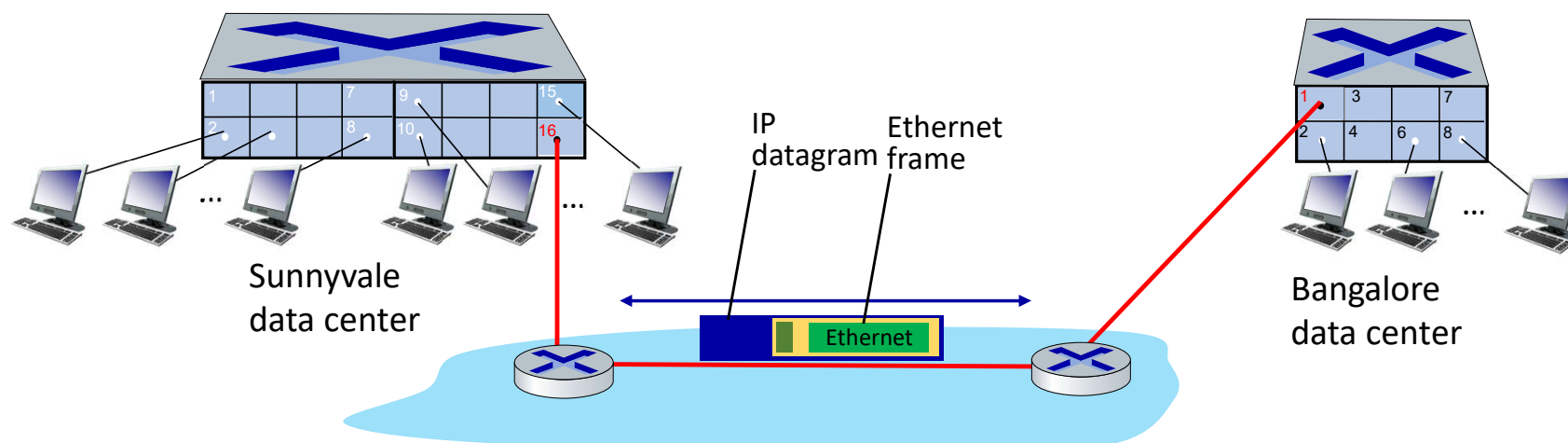
trunk port: carries frames between VLANs defined over multiple physical switches

- frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry VLAN ID info)
- 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

802.1Q VLAN frame format



EVPN: Ethernet VPNs (aka VXLANs)



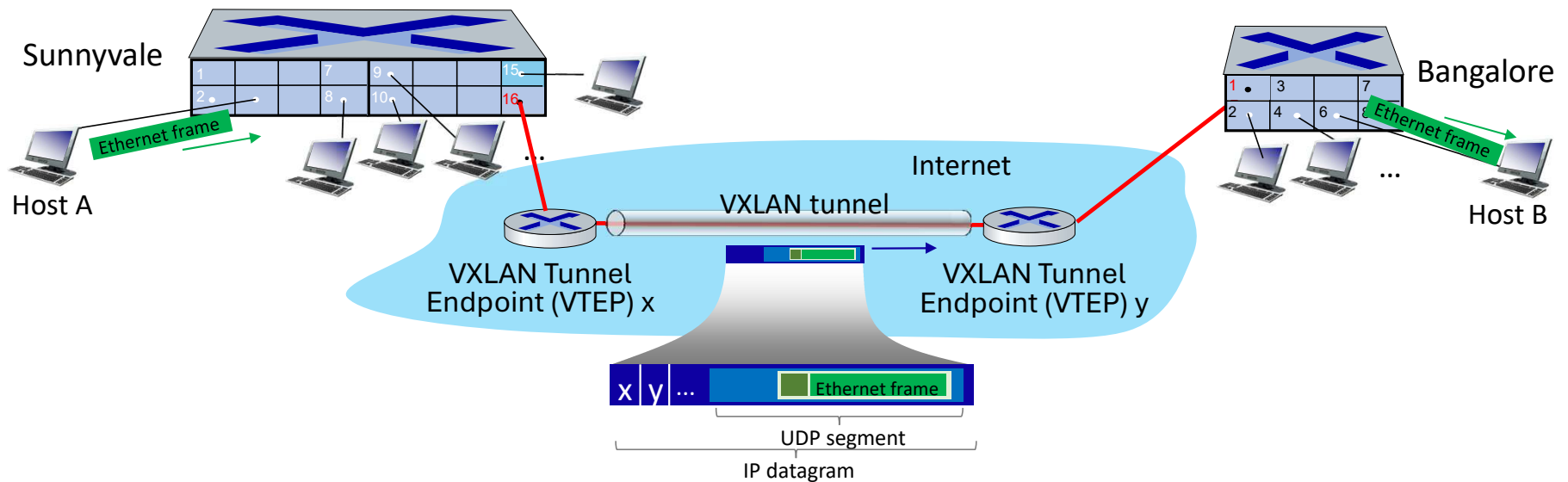
Layer-2 Ethernet switches *logically* connected to each other (e.g., using IP as an *underlay*)

- Ethernet frames carried *within* IP datagrams between sites
- “*tunneling*” scheme to *overlay Layer 2 networks on top of Layer 3 networks* ... runs over the existing networking infrastructure and provides a means to “stretch” a Layer 2 network.” [RFC 7348]

EVPN: Ethernet VPNs (aka VXLANs)

Layer-2 Ethernet switches *logically* connected to each other (using IP as an *underlay*)

- Ethernet frames carried *within* IP datagrams between sites
- “*tunneling*” scheme to *overlay Layer 2 networks on top of Layer 3 networks* ... runs over existing networking infrastructure ... provides a means to “stretch” a Layer 2 network.” [RFC 7348]



Link layer, LANs: roadmap

- LANs
 - addressing, ARP
 - switches
 - VLANs
- link virtualization: MPLS



♦ Evolution of the Concept of a “Link”

Initial View

- A link was first seen as a physical wire connecting two hosts.

As Understanding Grew

1. Multiple Access Networks

- Many hosts share one medium (Ethernet, Wi-Fi)
- Link becomes a shared channel, not just a wire

2. Switched LANs

- The “link” may be a complex switched infrastructure
- Hosts still *perceive* it as a single link-layer connection

3. VLANs

- Physical LAN split into logical LANs
- Hosts remain unaware of internal complexity

📌 Key idea:

Even if the underlying infrastructure is complex, hosts always see a simple link-layer channel.

♦ Virtualization of Networks

Dial-Up Example

- Dial-up modem uses the telephone network
- Telephone network has:
 - Its own switches
 - Its own protocols
- Yet the Internet treats it as a simple link

➡ Internet virtualizes lower networks

Overlay Analogy

- Internet overlays the telephone network
- Overlay networks overlay the Internet
- Each layer treats the lower one as a “link”

♦ Why MPLS Fits Here

- MPLS is:
 - Packet-switched
 - Virtual-circuit-based
- Can be viewed as:
 - Network layer ❌
 - Link layer ✅ (from IP’s point of view)

📌 MPLS interconnects IP devices, just like Ethernet or ATM

♦ What Is MPLS?

Multiprotocol Label Switching (MPLS) is a technique that:

- Combines IP routing with virtual-circuit switching
- Uses short fixed-length labels instead of IP lookups

📌 Goal:

- Improve forwarding flexibility
- Enable advanced traffic engineering
- Not to replace IP, but to augment it

♦ MPLS Header Placement

An MPLS frame looks like this:

less

📄 Copy code

| Ethernet Header | MPLS Header | IP Header | Payload |

MPLS Header Fields

Field	Purpose
Label	Identifies the forwarding path
Exp (3 bits)	QoS / experimental
S bit	End of label stack
TTL	Prevents infinite loops

📌 MPLS header sits between Layer 2 and Layer 3

♦ MPLS-Capable Routers

- Called Label-Switched Routers (LSRs)
- Forward packets by:
 - Looking up label
 - NOT examining IP address

📌 Faster and more flexible forwarding

⚠ MPLS frames can only be exchanged between MPLS-capable routers

How MPLS Forwarding Works (Figure 6.29 Concept)

- Routers advertise:
 - Which destinations they can reach
 - Which label corresponds to that destination

Example

- Router R4 has two MPLS paths to destination A
- Each path uses a different label and interface

- ✚ MPLS routers never touch the IP header
- ✚ MPLS behaves like a virtual circuit network

♦ Label Distribution

- Labels are distributed using signaling protocols
- Common choice:
 - RSVP-TE (extension of RSVP)

Routing info (bandwidth, link state) is:

- Flooded using extensions of OSPF

⚠ Path computation is vendor-specific

♦ Why MPLS Is Important (Real Advantage)

- ⊘ Speed is NOT the main benefit anymore
- ✅ Traffic Engineering is

♦ Traffic Engineering with MPLS

Traditional IP Routing

- Only one least-cost path
- No control over traffic distribution

MPLS

- Multiple paths to same destination
- Network operator can:
 - Split traffic
 - Enforce policies
 - Avoid congestion

✚ MPLS allows non-shortest paths on purpose

♦ Additional Uses of MPLS

1 Fast Failure Recovery

- Precomputed backup paths
- Quick rerouting after link failure

2 Virtual Private Networks (VPNs)

- ISPs use MPLS to:
 - Isolate customer traffic
 - Maintain private addressing
- Common in enterprise WANs

♦ MPLS vs SDN

- MPLS came before SDN
- Many MPLS features can now be done with:
 - SDN
 - Generalized forwarding

✚ Open question:

Will MPLS and SDN coexist, or will SDN replace MPLS?

📝 One-Line Exam Answers

What is MPLS?

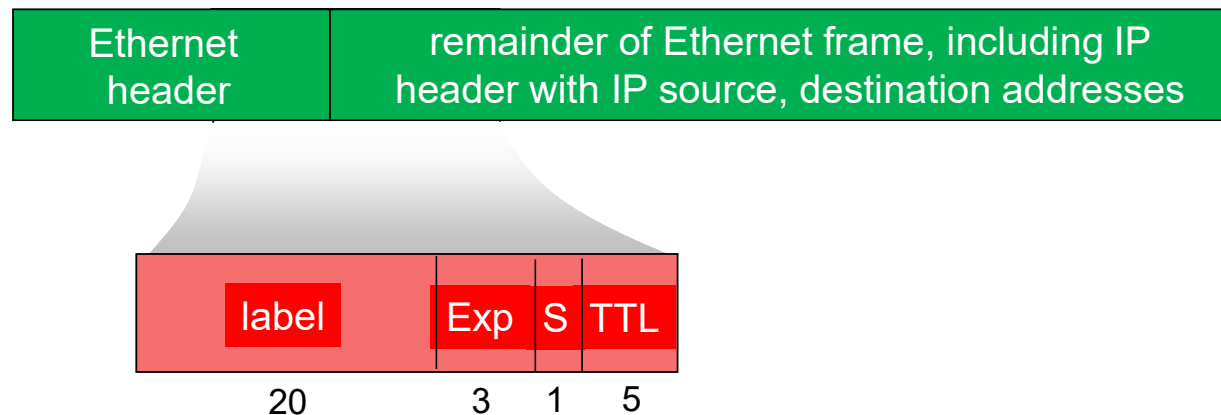
MPLS is a packet-switched, virtual-circuit technology that forwards packets using short labels instead of IP addresses, enabling traffic engineering and flexible routing.

Why MPLS is considered link-layer?

From IP's perspective, MPLS acts as a link-layer technology that interconnects IP devices without examining IP headers.

Multiprotocol label switching (MPLS)

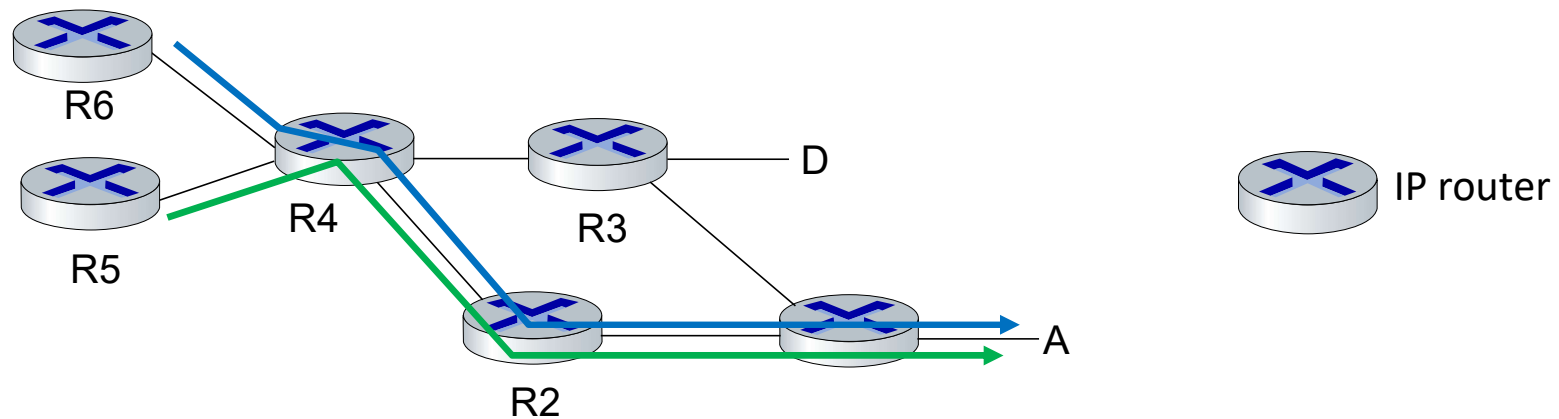
- **goal:** high-speed IP forwarding among network of MPLS-capable routers, using fixed length label (instead of shortest prefix matching)
 - faster lookup using fixed length identifier
 - borrowing ideas from Virtual Circuit (VC) approach
 - but IP datagram still keeps IP address!



MPLS capable routers

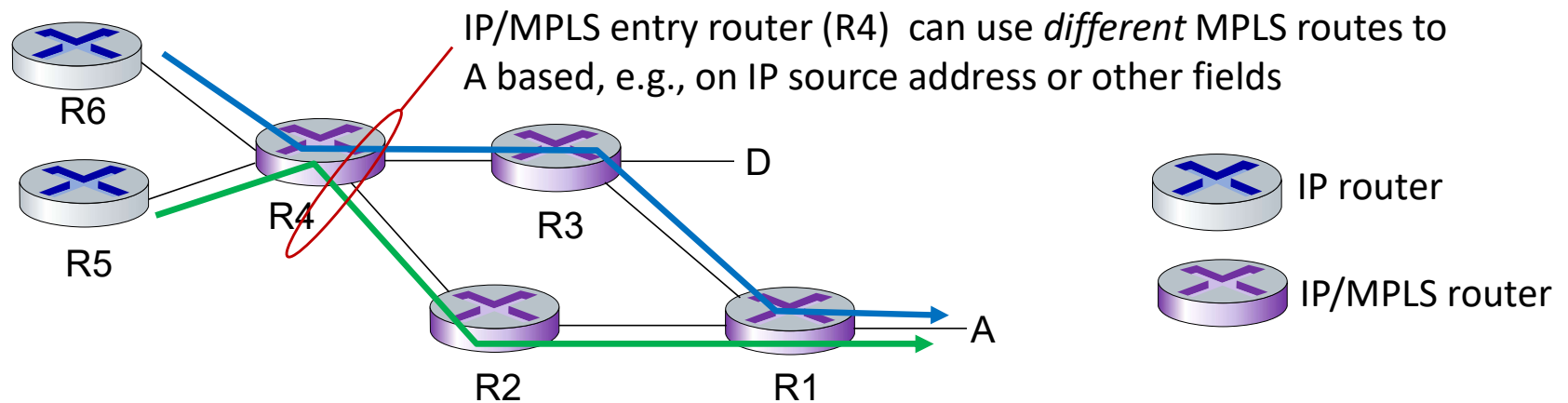
- a.k.a. label-switched router
- forward packets to outgoing interface based only on label value (*don't inspect IP address*)
 - MPLS forwarding table distinct from IP forwarding tables
- *flexibility*: MPLS forwarding decisions can *differ* from those of IP
 - use destination *and* source addresses to route flows to same destination differently (traffic engineering)
 - re-route flows quickly if link fails: pre-computed backup paths

MPLS versus IP paths



- **IP routing:** path to destination determined by destination address alone

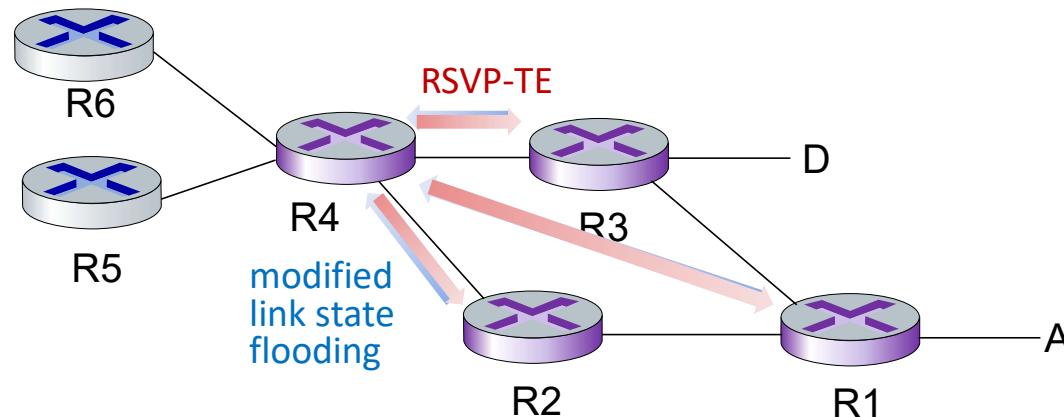
MPLS versus IP paths



- **IP routing:** path to destination determined by destination address alone
- **MPLS routing:** path to destination can be based on source *and* destination address
 - flavor of generalized forwarding (MPLS 10 years earlier)
 - *fast reroute*: precompute backup routes in case of link failure

MPLS signaling

- modify OSPF, IS-IS link-state flooding protocols to carry info used by MPLS routing:
 - e.g., link bandwidth, amount of “reserved” link bandwidth
- entry MPLS router uses RSVP-TE signaling protocol to set up MPLS forwarding at downstream routers



MPLS forwarding tables

